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Chen et al.

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(54) SUSPEND-RESUME-GO TECHNIQUES FOR MEMORY DEVICES

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G06F 3/00 (2006.01)
G06F 3/06 (2006.01)

(52) **U.S. Cl.**
CPC ***G06F 3/0659*** (2013.01); ***G06F 3/0604*** (2013.01); ***G06F 3/0679*** (2013.01)

(58) **Field of Classification Search**
CPC G06F 3/0659; G06F 3/0604; G06F 3/0679
See application file for complete search history.

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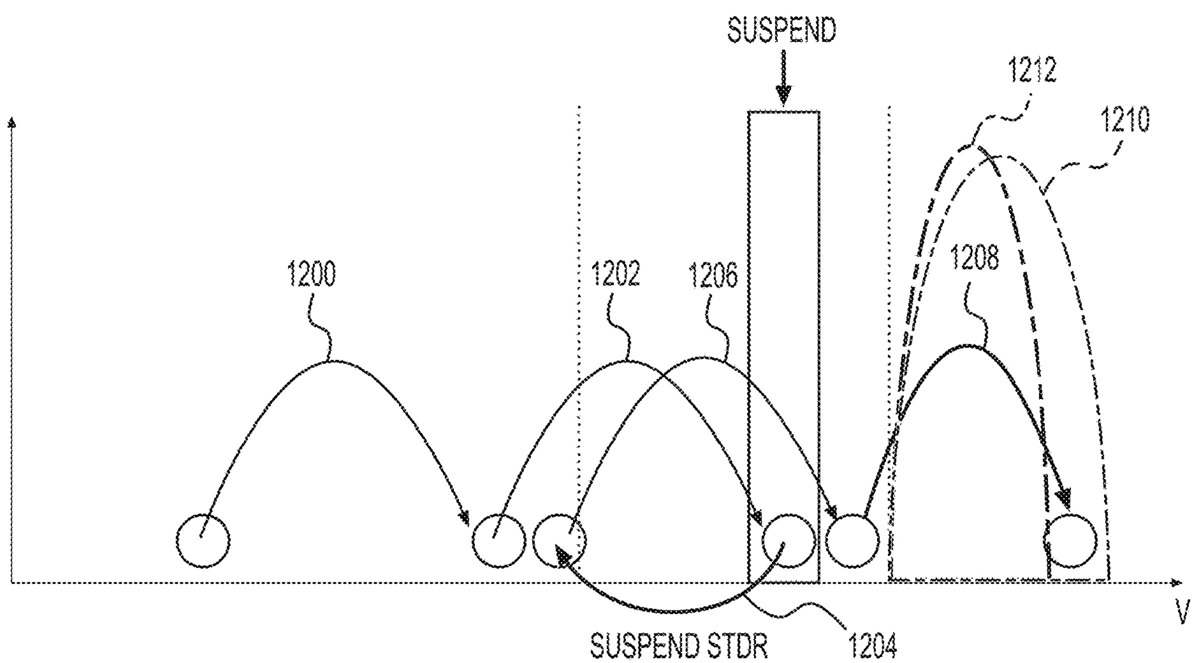
(57) **ABSTRACT**

The memory device includes a memory block with an array of memory cells that are arranged in word lines. The word lines are in electrical communication with respective word line drivers and switches. Control circuitry is configured to program the memory cells of a selected word line in a programming operation during which the control circuitry applies an elevated voltage to the selected word line and receives a command to suspend the programming operation. With the word line switch associated with the selected word line turned on, the control circuitry ramps the selected word line from the elevated voltage to a reduced gate holding voltage and then turn the word line switch associated with the selected word line off to electrically isolate the selected word line from the associated word line driver so that the selected word line remains at the gate holding voltage until the programming operation resumes.

20 Claims, 19 Drawing Sheets



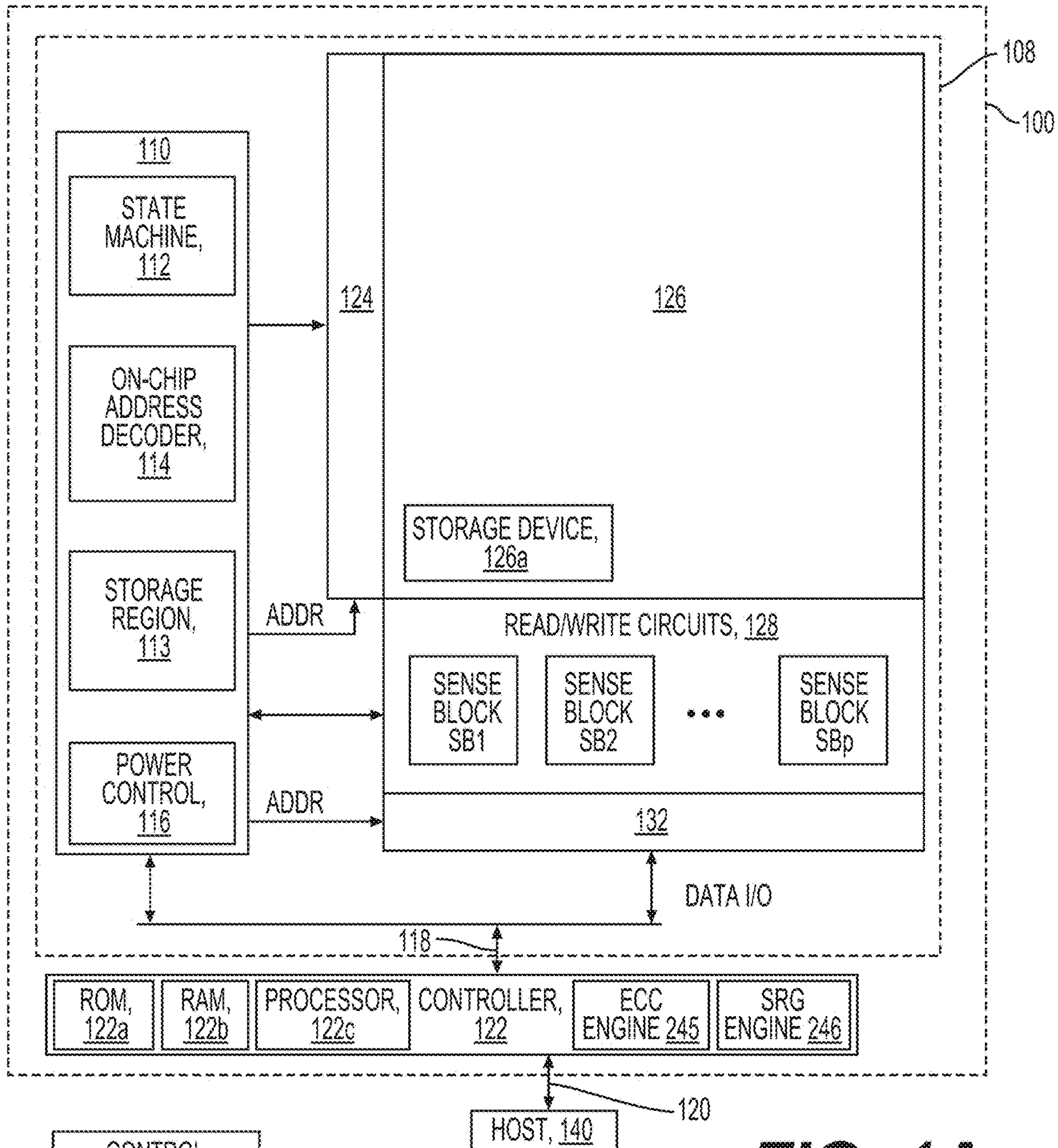

108
100
110
STATE MACHINE, 112
ON-CHIP ADDRESS DECODER, 114
STORAGE REGION, 113
POWER CONTROL, 116
124
126
STORAGE DEVICE, 126a
ADDR
READ/WRITE CIRCUITS, 128
SENSE BLOCK SB1
SENSE BLOCK SB2
SENSE BLOCK SBp
ADDR
132
DATA I/O
118
ROM, 122a
RAM, 122b
PROCESSOR, 122c
CONTROLLER, 122
ECC ENGINE 245
SRG ENGINE 246
120
HOST, 140

FIG. 1A

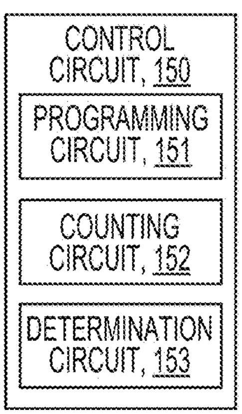

CONTROL CIRCUIT, 150
PROGRAMMING CIRCUIT, 151
COUNTING CIRCUIT, 152
DETERMINATION CIRCUIT, 153

FIG. 1B

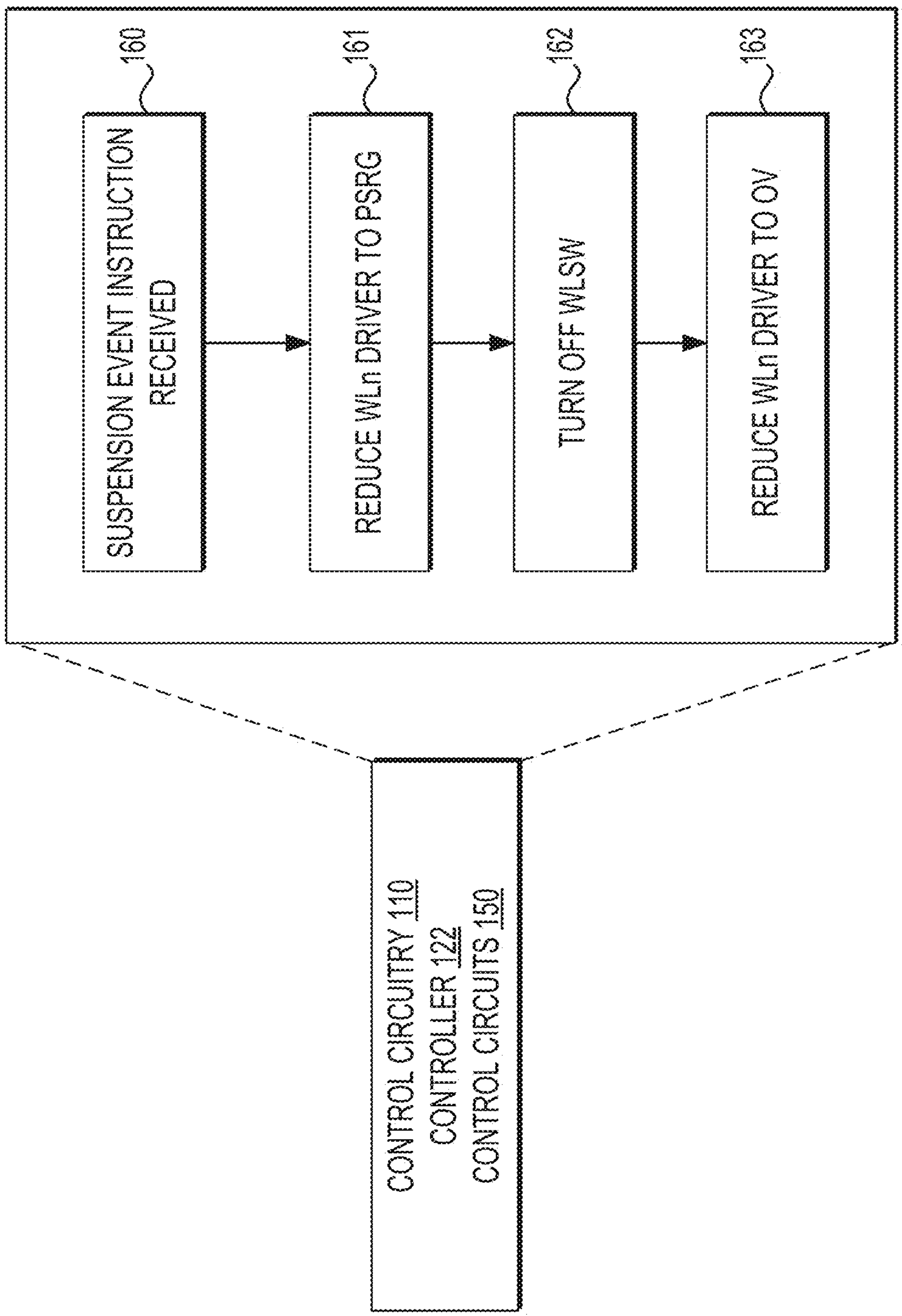

160
SUSPENSION EVENT INSTRUCTION RECEIVED
161
REDUCE WLn DRIVER TO PSRG
162
TURN OFF WLSW
163
REDUCE WLn DRIVER TO 0V
CONTROL CIRCUITRY 110
CONTROLLER 122
CONTROL CIRCUITS 150

FIG. 1C

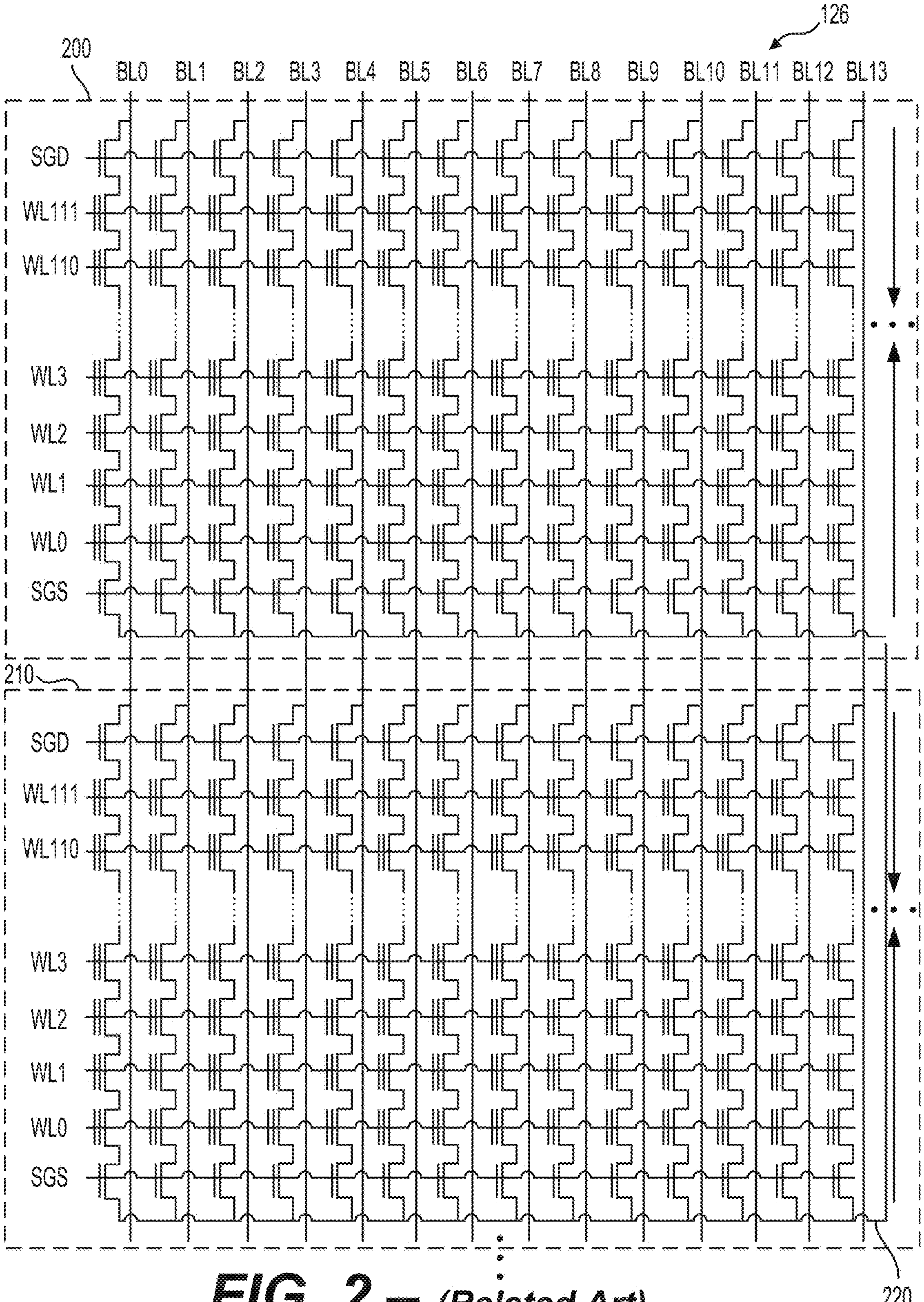


FIG. 2 — (Related Art)

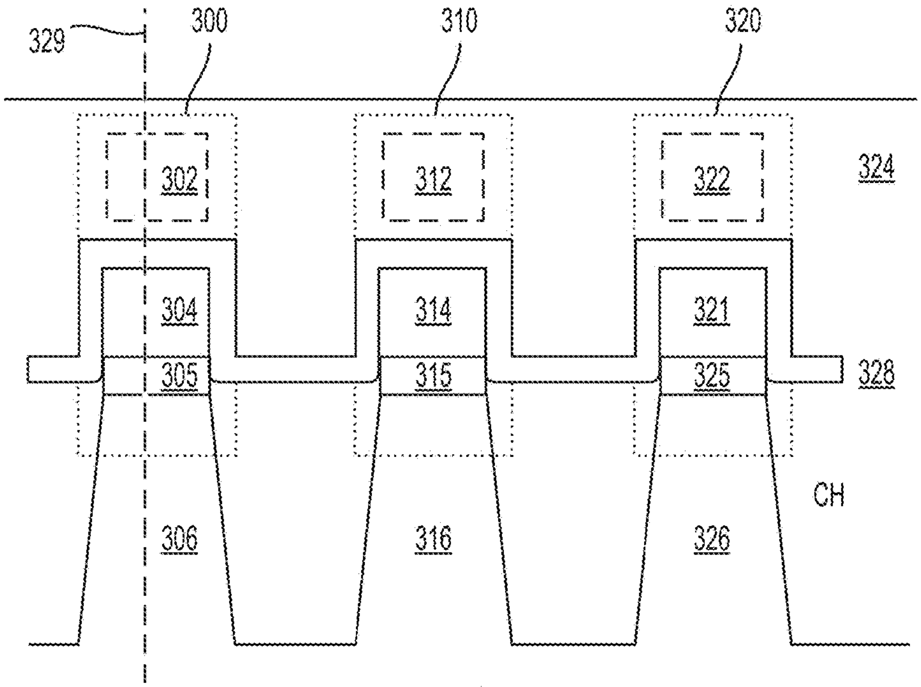


FIG. 3A
(Related Art)

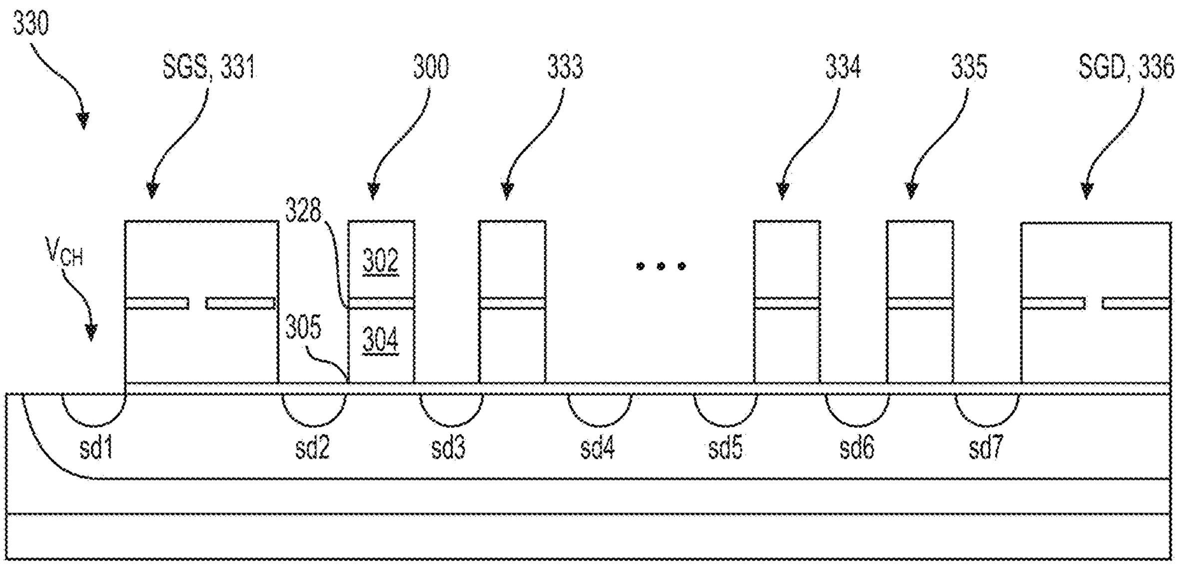


FIG. 3B
(Related Art)

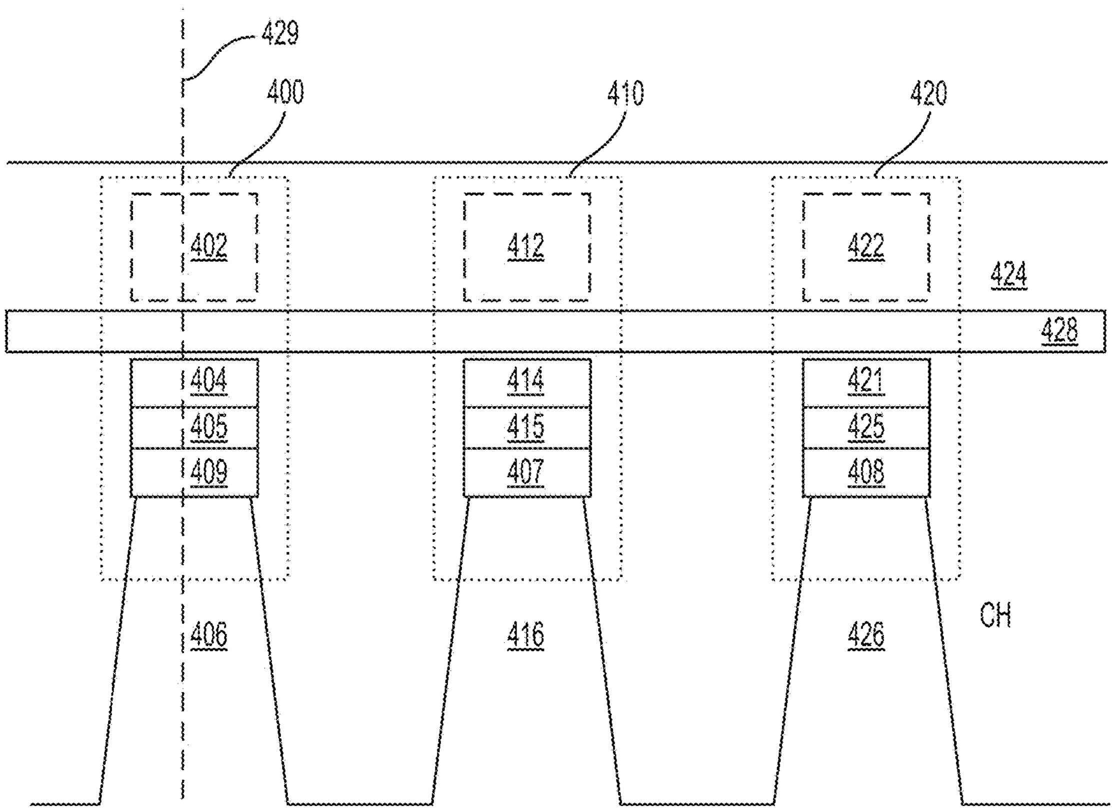


FIG. 4A
(Related Art)

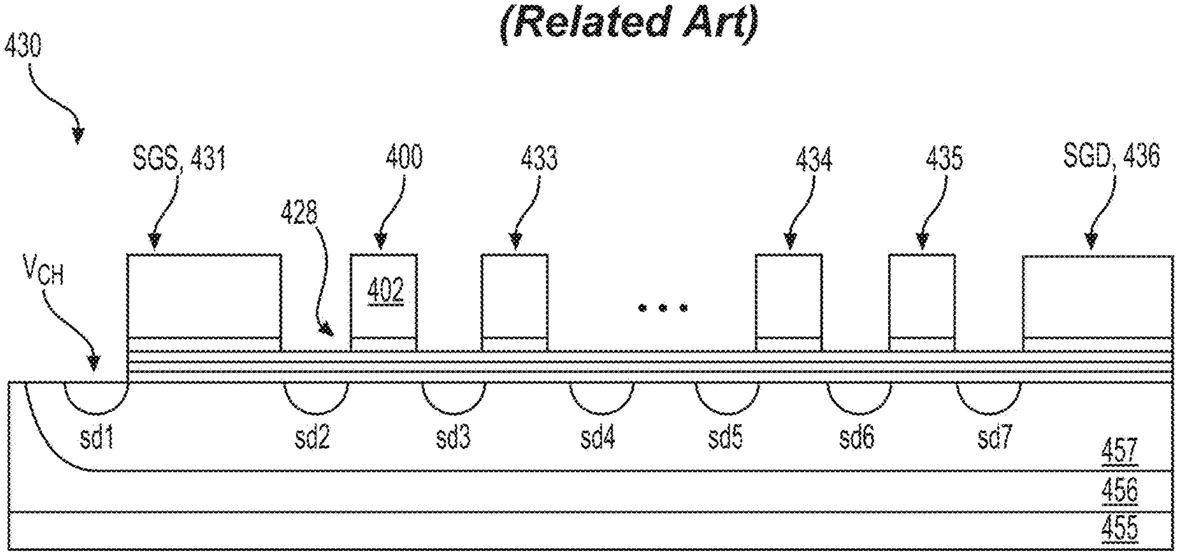


FIG. 4B
(Related Art)

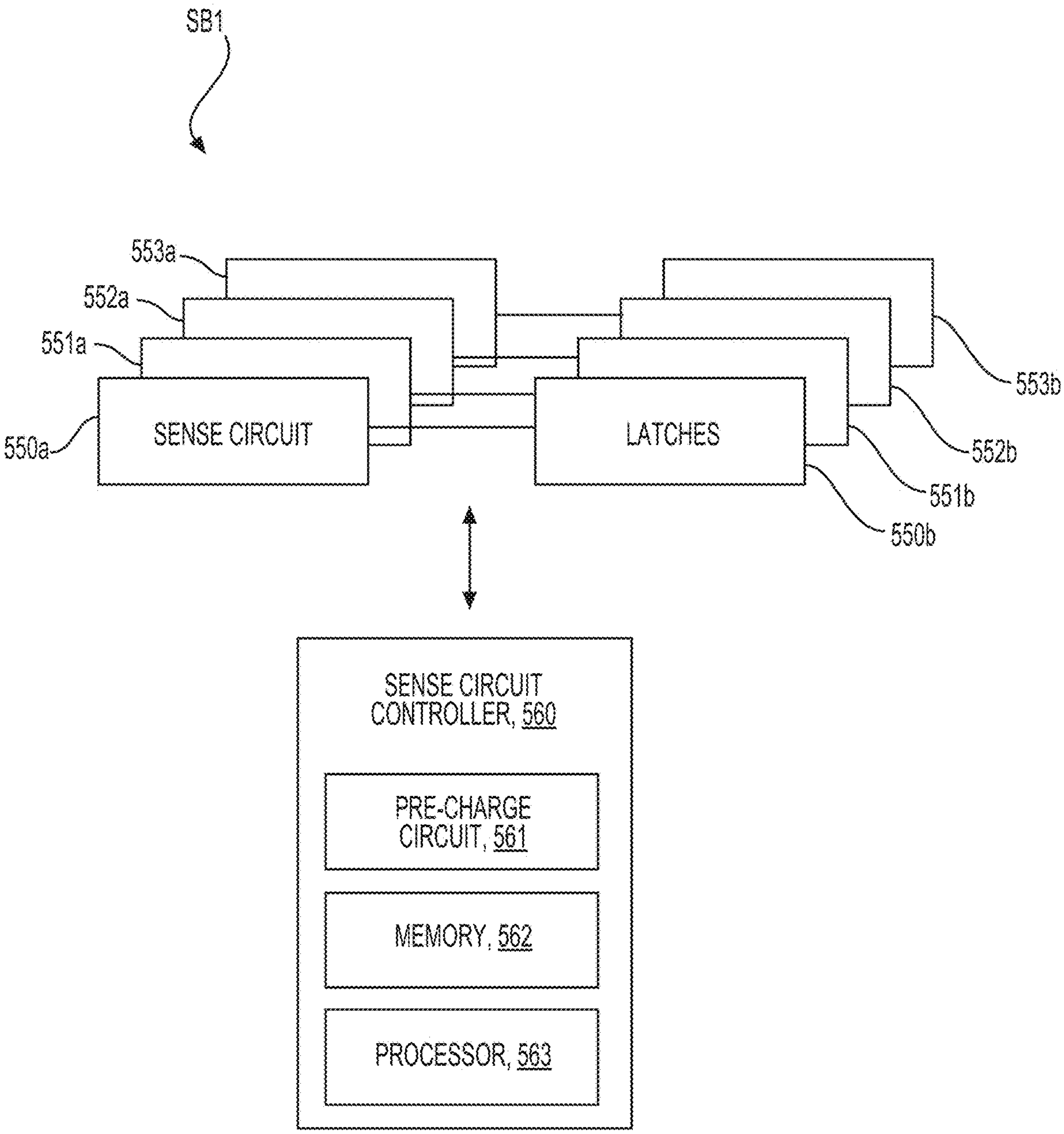

SB1
553a
552a
551a
550a
SENSE CIRCUIT
LATCHES
553b
552b
551b
550b
SENSE CIRCUIT CONTROLLER, 560
PRE-CHARGE CIRCUIT, 561
MEMORY, 562
PROCESSOR, 563

FIG. 5

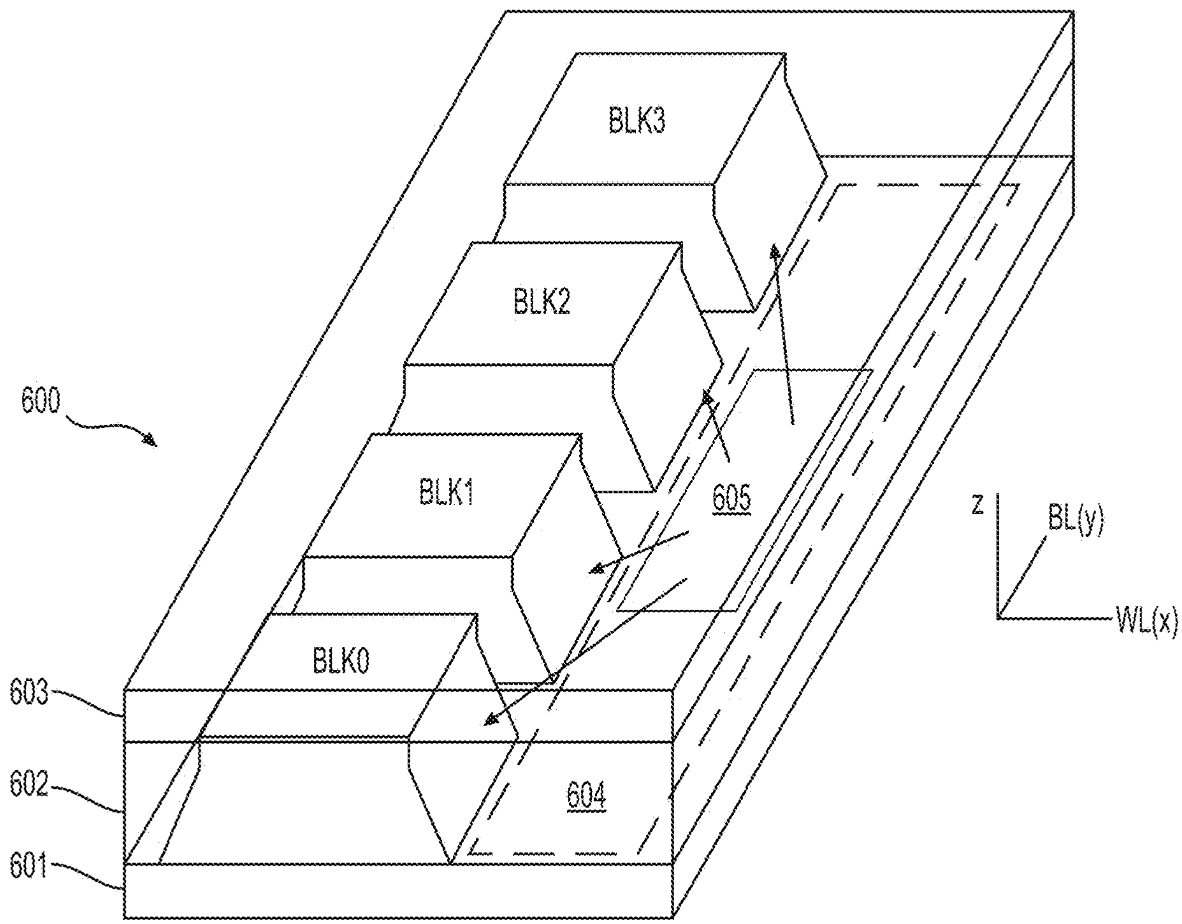


FIG. 6A
(Related Art)

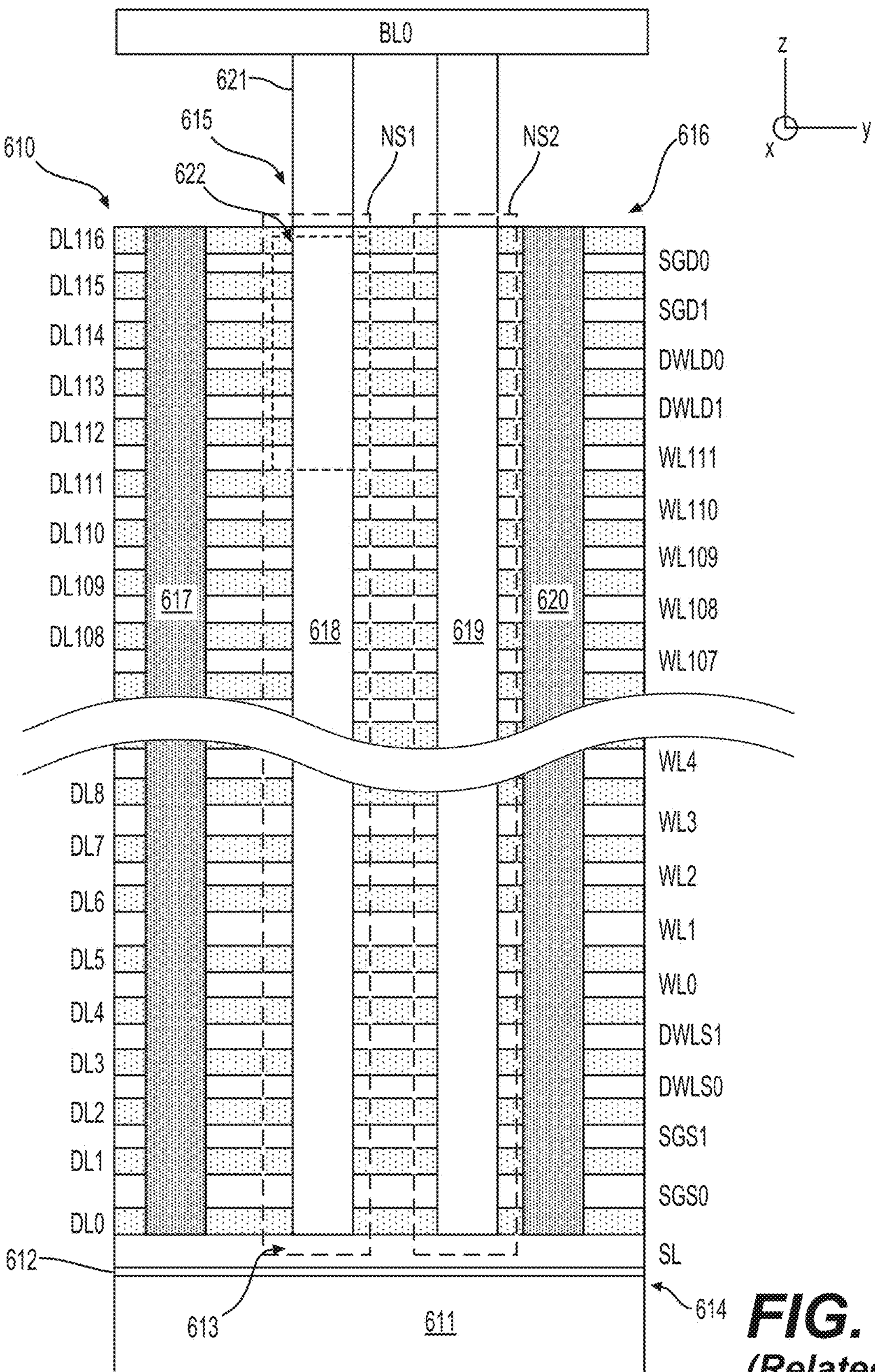


FIG. 6B
(Related Art)

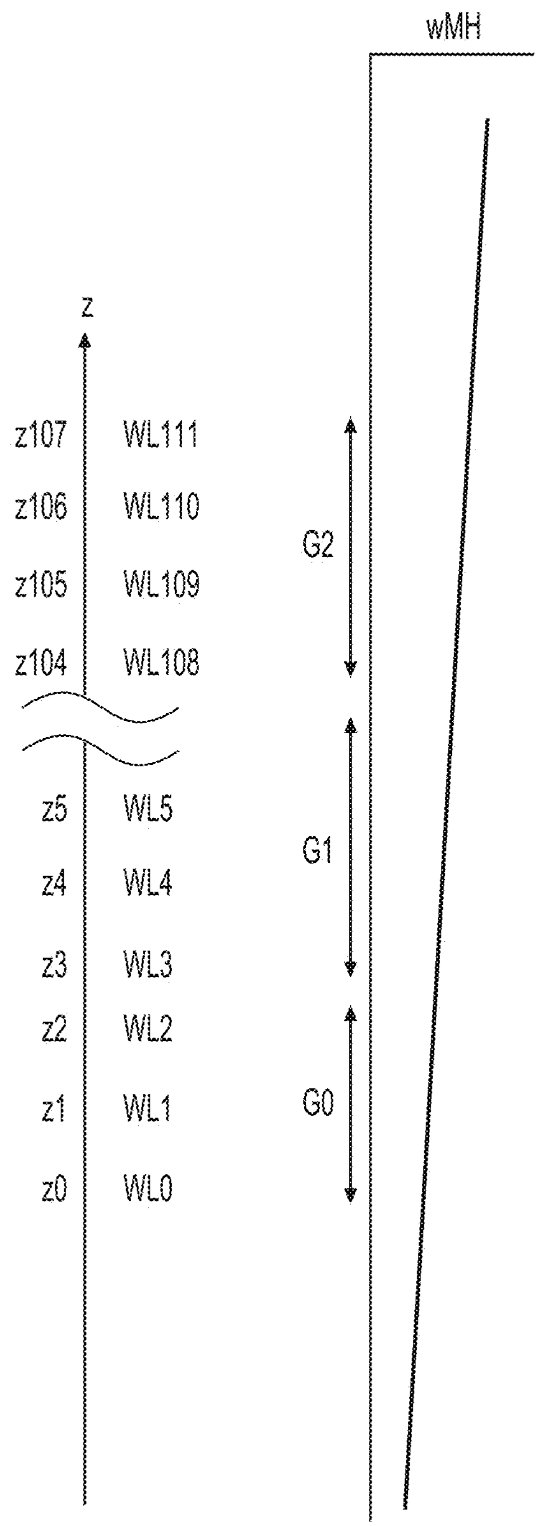


FIG. 6C
(Related Art)

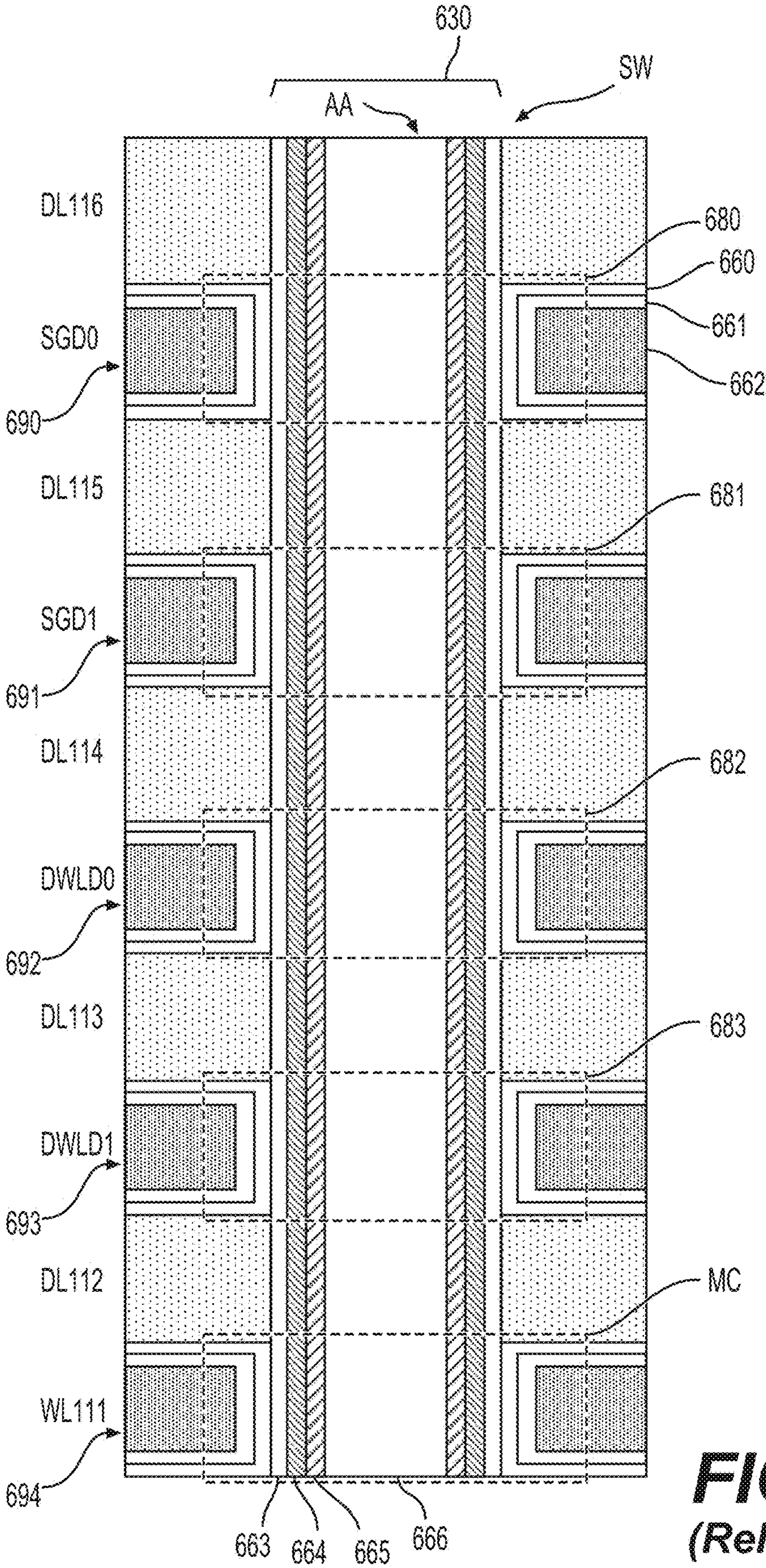


FIG. 6D
(Related Art)

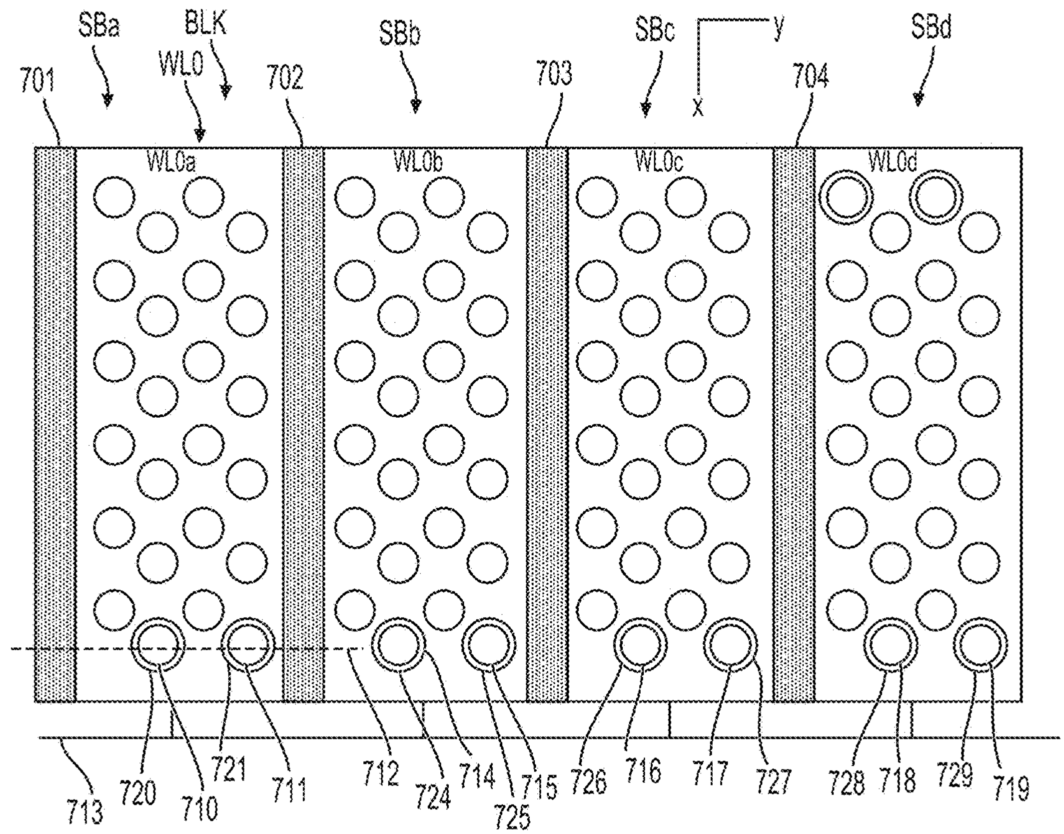


FIG. 7A – (Related Art)

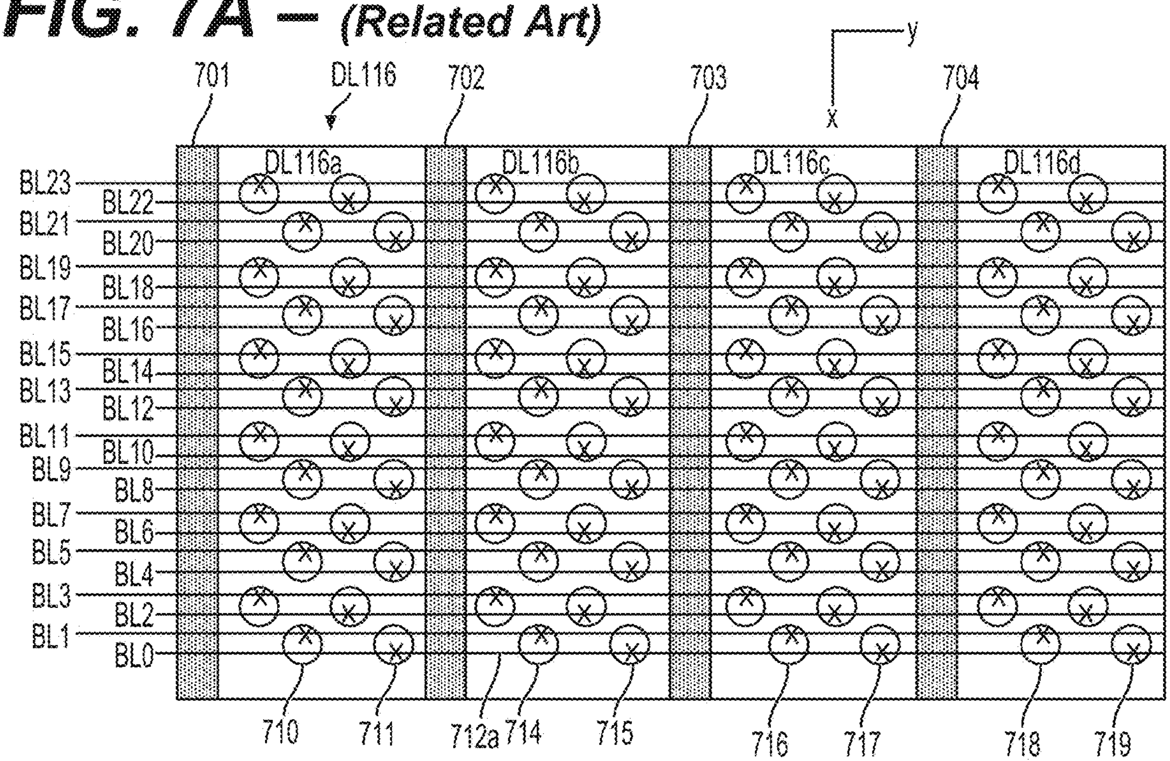
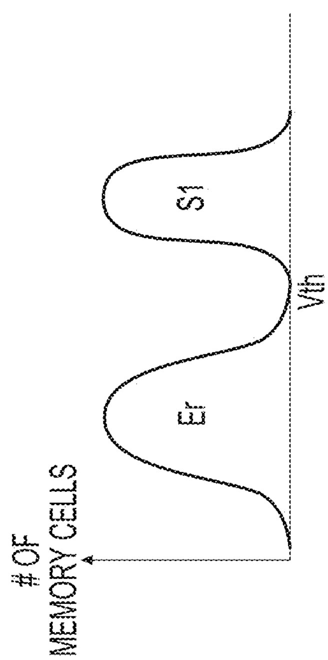
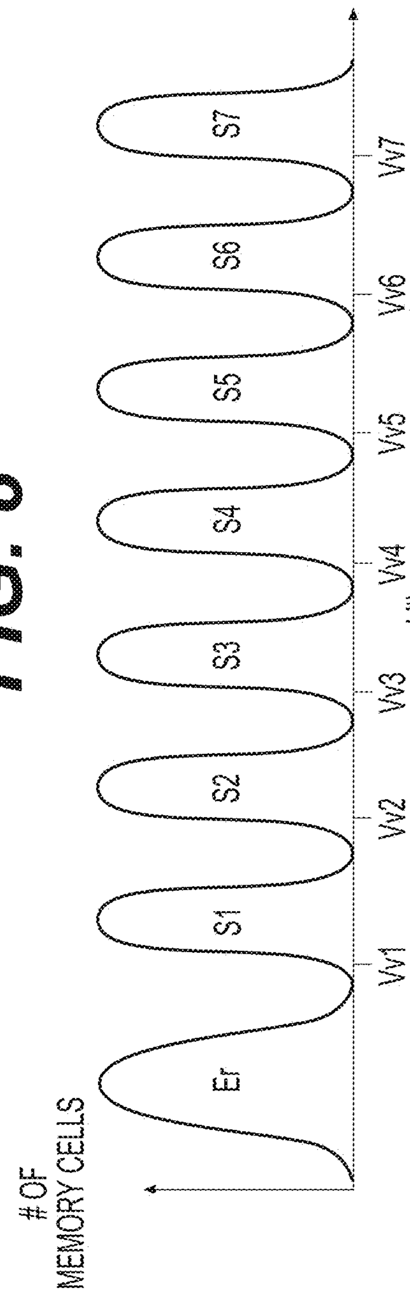


FIG. 7B - (Related Art)



OF
MEMORY CELLS
Er
S1
Vth

FIG. 8



OF
MEMORY CELLS
Er
S1
S2
S3
S4
S5
S6
S7
Vv1
Vv2
Vv3
Vv4
Vv5
Vv6
Vv7
Vth

FIG. 9

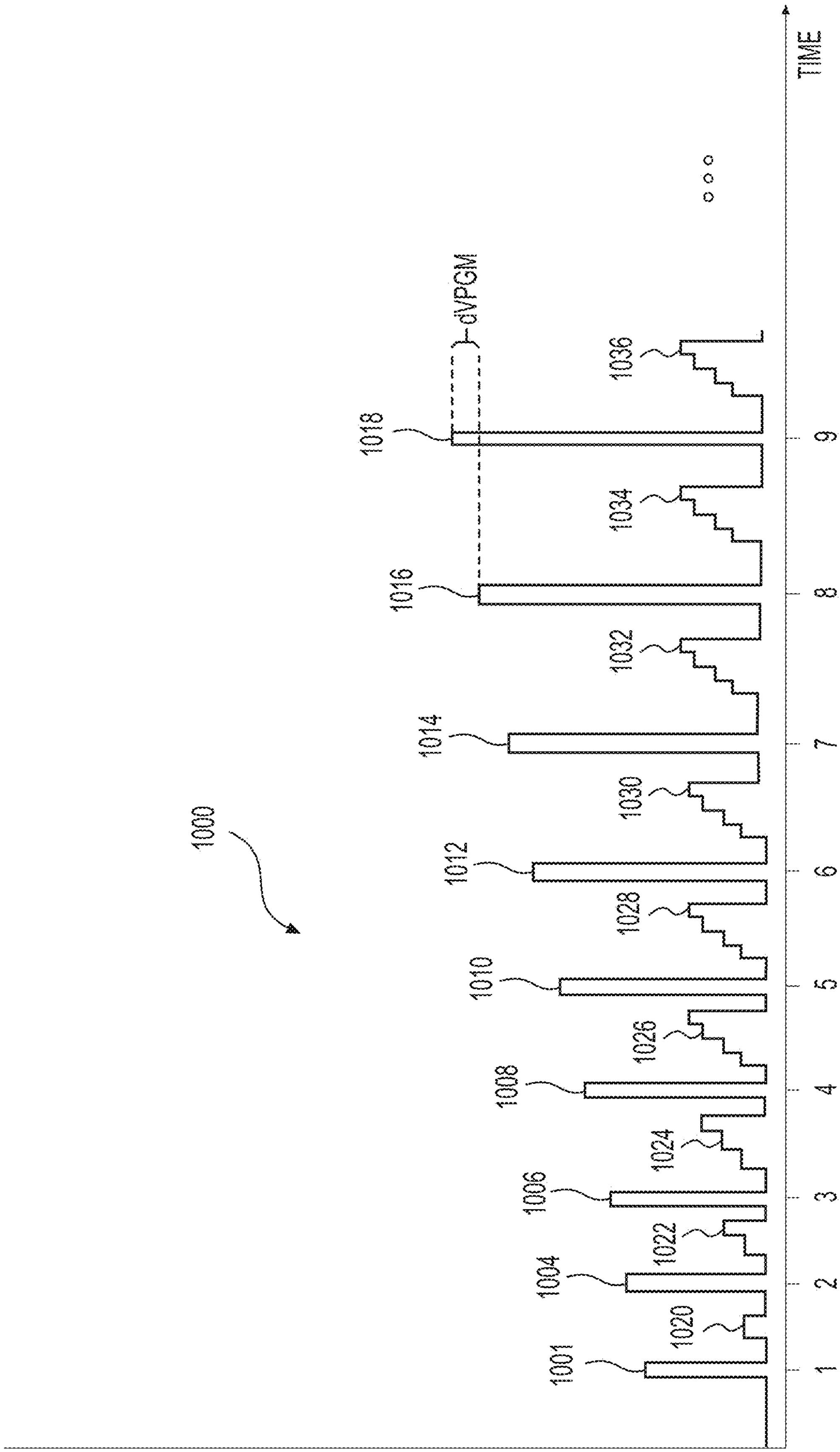

1000
1001
1004
1006
1008
1010
1012
1014
1016
1018
1020
1022
1024
1026
1028
1030
1032
1034
1036
dVPGM
TIME
1
2
3
4
5
6
7
8
9

FIG. 10

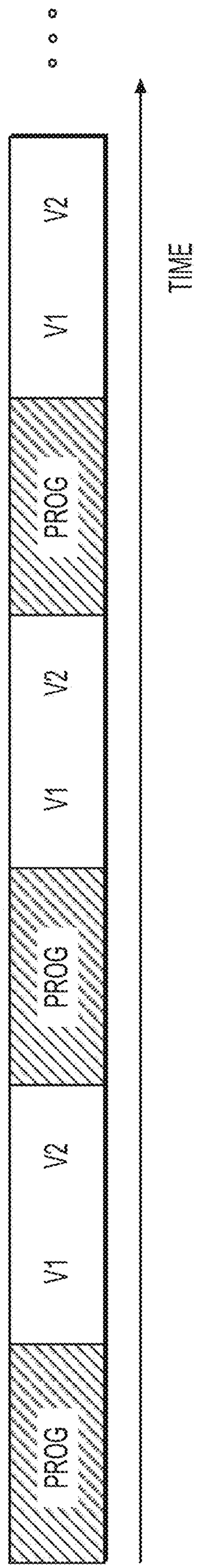

PROG
V1
V2
PROG
V1
V2
PROG
V1
V2
TIME

FIG. 11

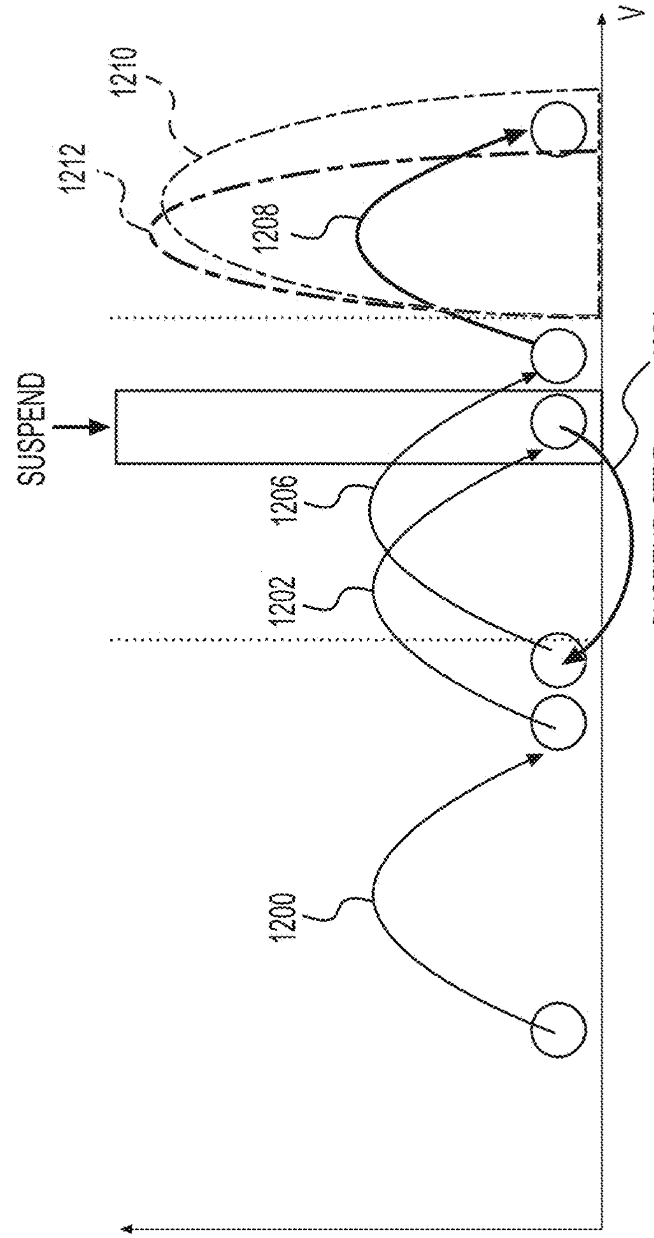

V
1210
1212
1208
SUSPEND
1204
1206
1202
SUSPEND STDR
1200

FIG. 12

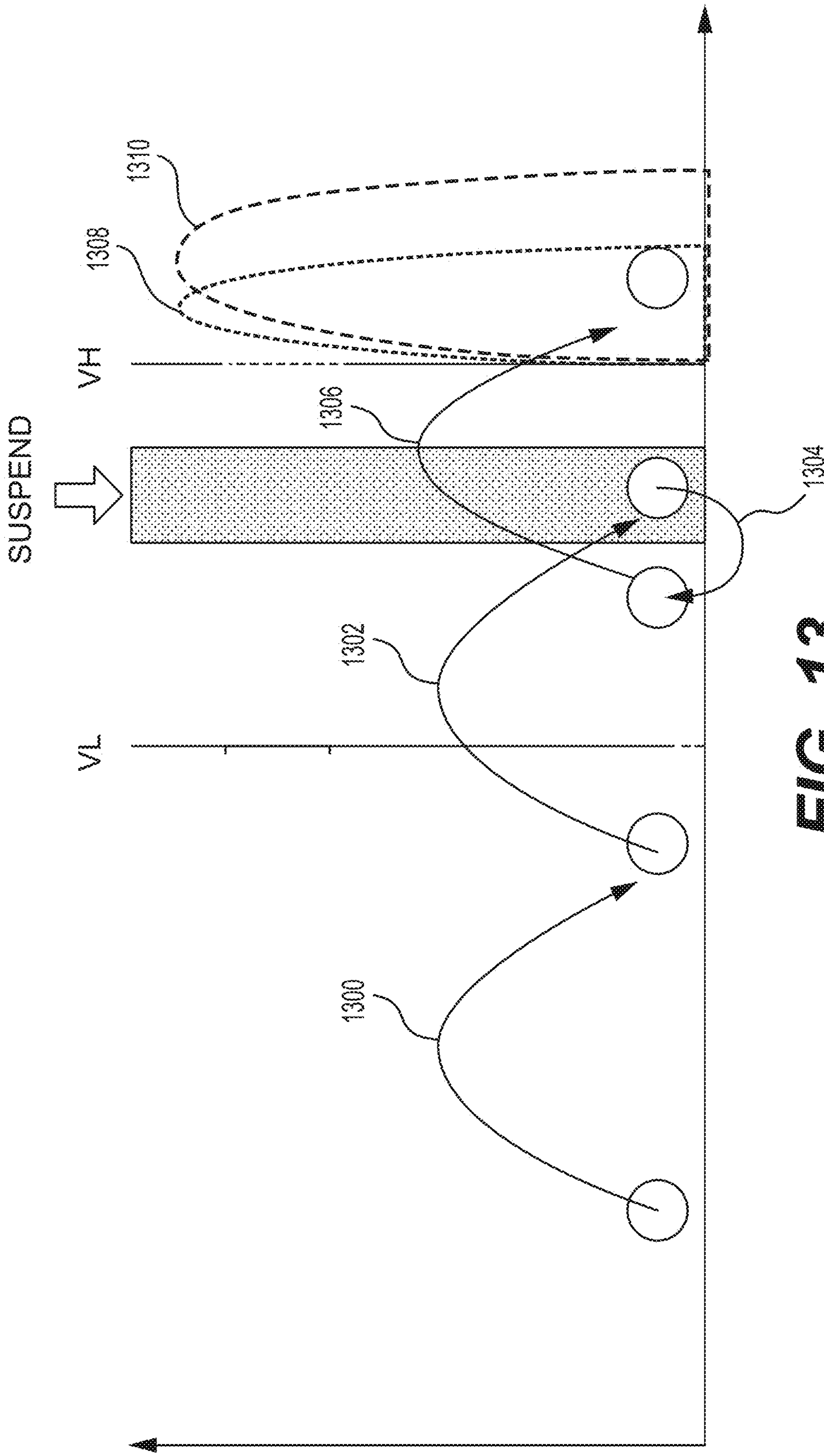


FIG. 13

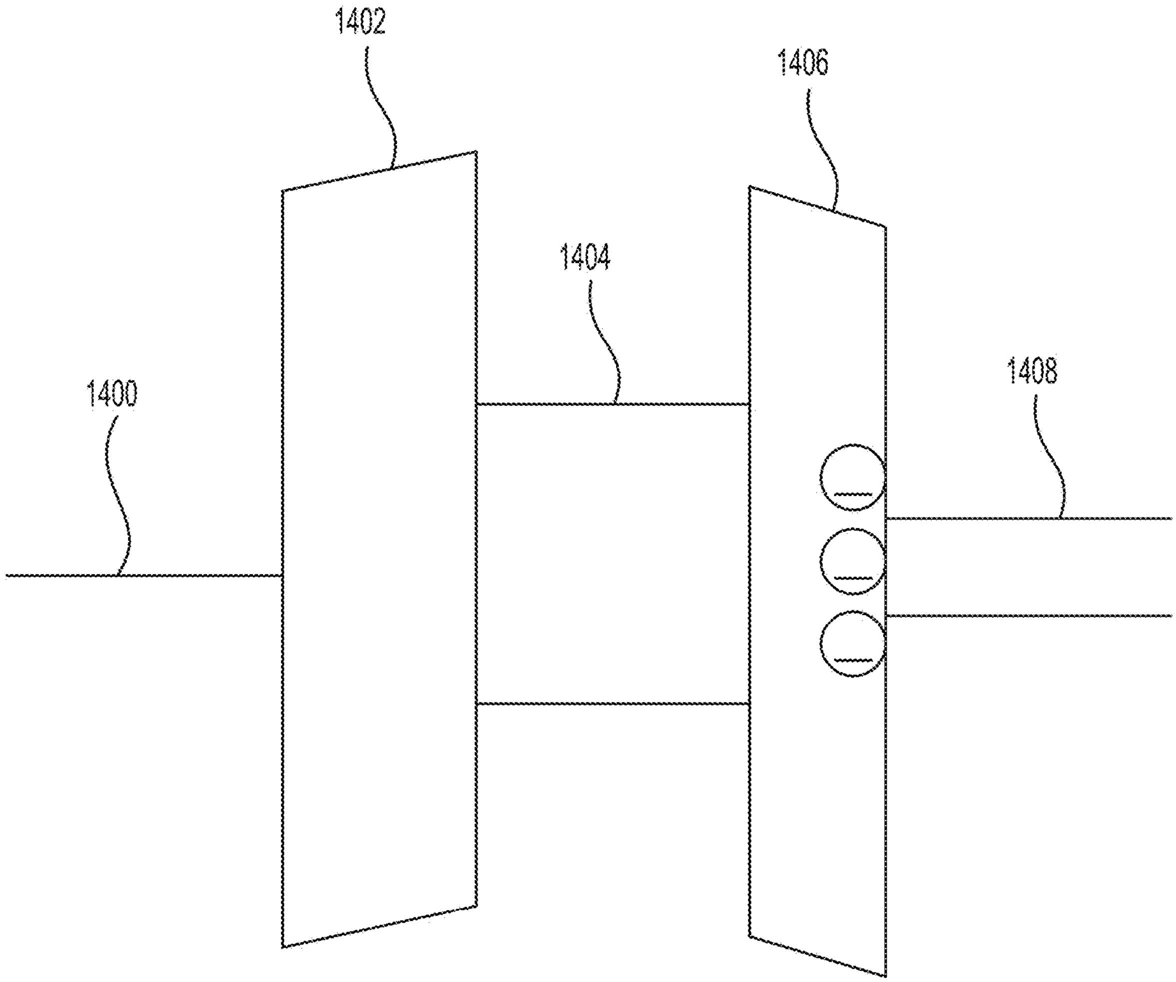


FIG. 14

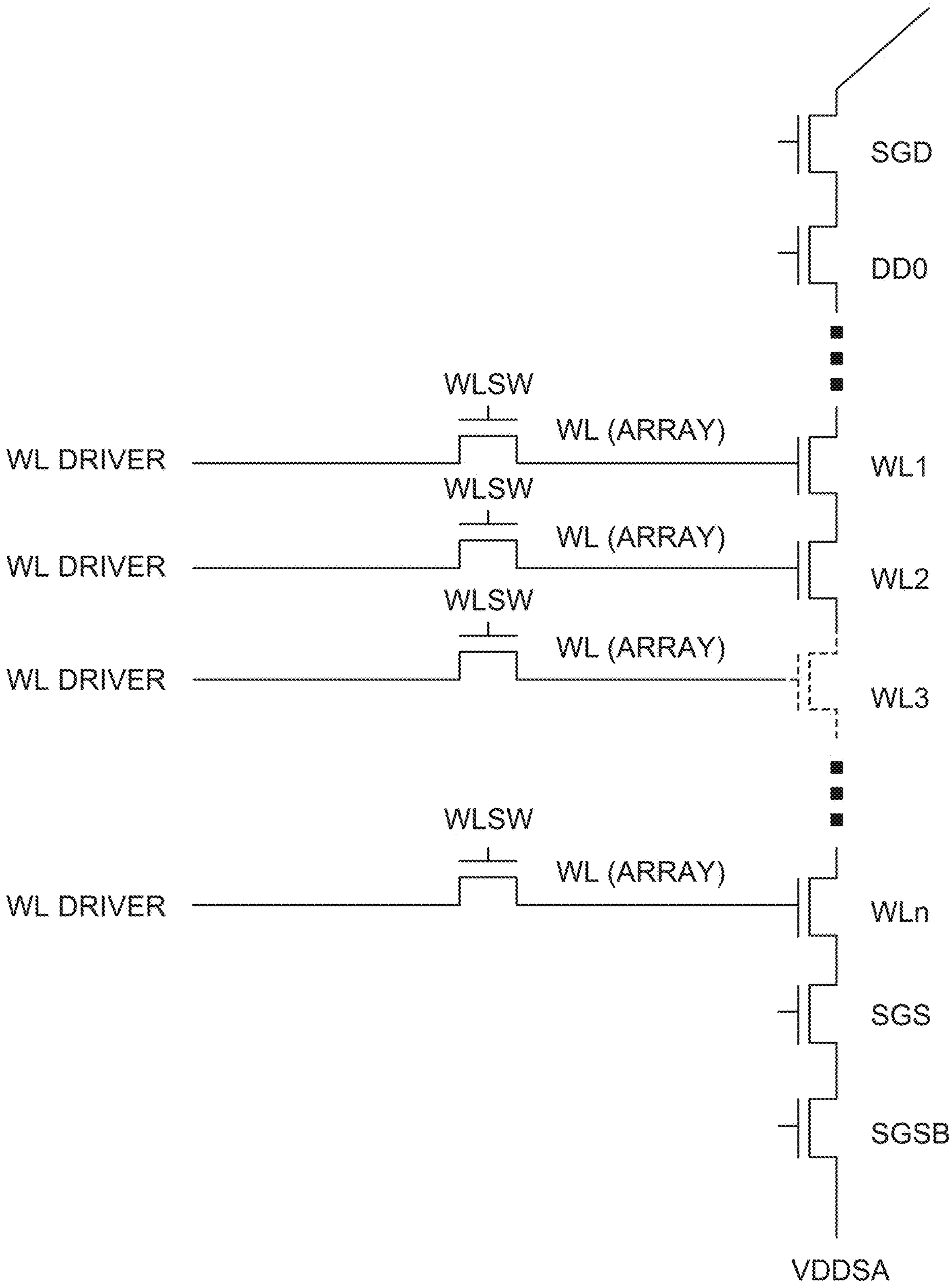

SGD
DD0
WLSW
WL (ARRAY)
WL DRIVER
WL1
WLSW
WL (ARRAY)
WL DRIVER
WL2
WLSW
WL (ARRAY)
WL DRIVER
WL3
WLSW
WL (ARRAY)
WL DRIVER
WLn
SGS
SGSB
VDDSA

FIG. 15

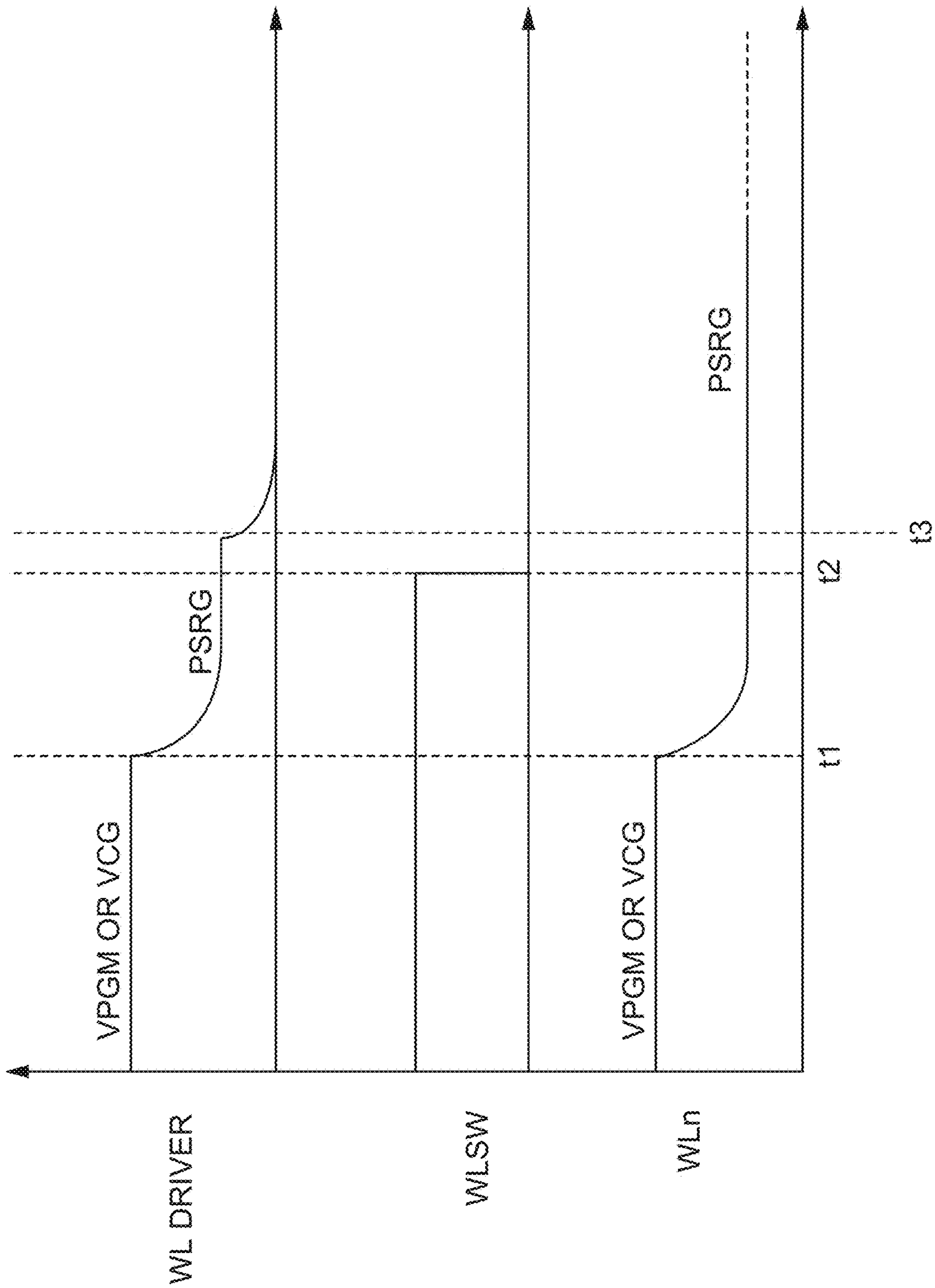


FIG. 16

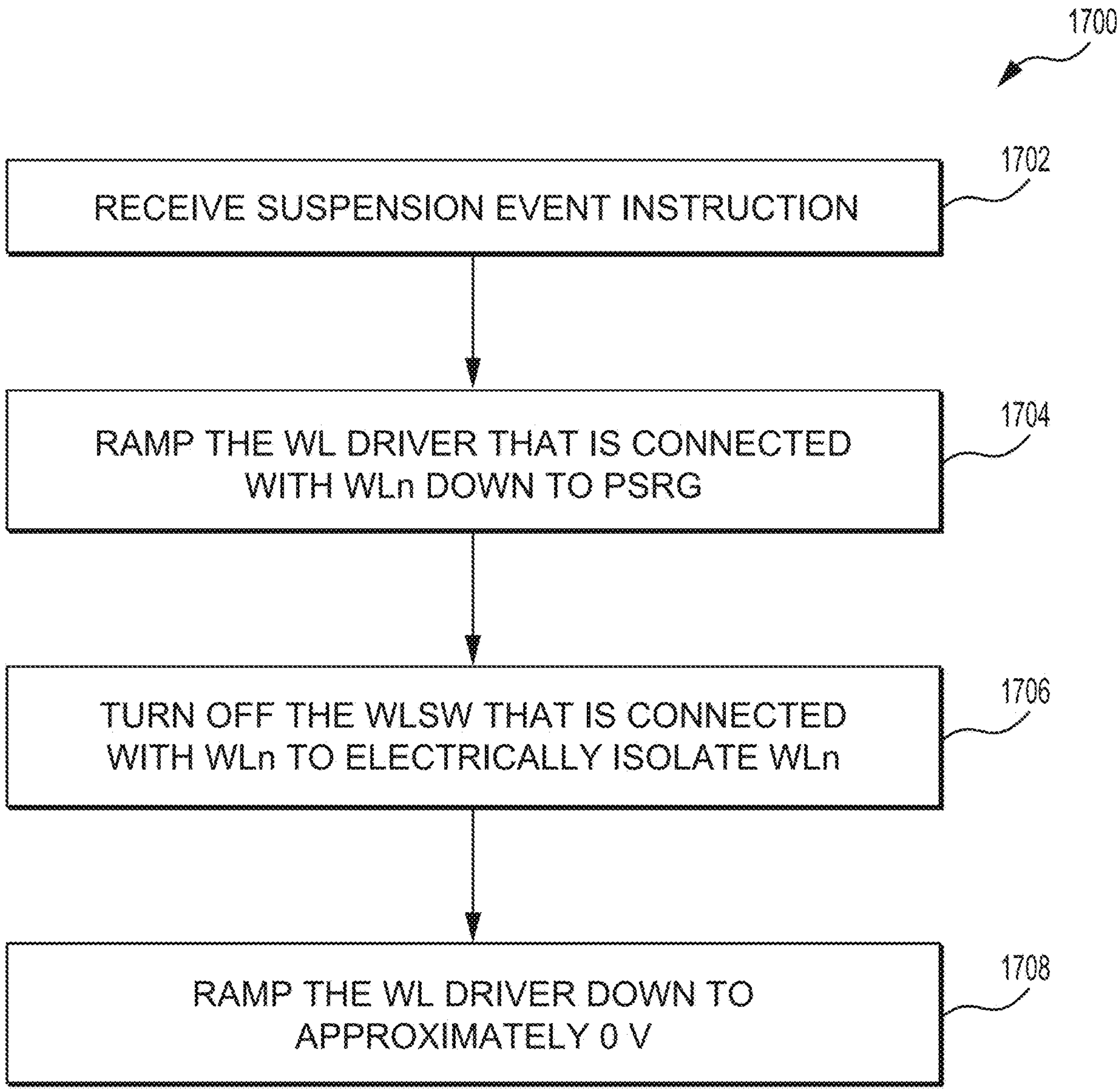

1700
1702
RECEIVE SUSPENSION EVENT INSTRUCTION
1704
RAMP THE WL DRIVER THAT IS CONNECTED WITH WLn DOWN TO PSRG
1706
TURN OFF THE WLSW THAT IS CONNECTED WITH WLn TO ELECTRICALLY ISOLATE WLn
1708
RAMP THE WL DRIVER DOWN TO APPROXIMATELY 0 V

FIG. 17

SUSPEND-RESUME-GO TECHNIQUES FOR MEMORY DEVICES

BACKGROUND

1. Field

The subject disclosure is related generally to techniques for programming memory cells in word lines and, more particularly, to techniques for improving the programming of the memory cells after a suspension occurs.

2. Related Art

Semiconductor memory is widely used in various electronic devices, such as cellular telephones, digital cameras, personal digital assistants, medical electronics, mobile computing devices, servers, solid state drives, non-mobile computing devices and other devices. Semiconductor memory may comprise non-volatile memory or volatile memory. A non-volatile memory allows information to be stored and retained even when the non-volatile memory is not connected to a source of power, e.g., a battery.

NAND memory devices include a chip with a plurality of memory blocks, each of which includes an array of memory cells arranged in a plurality of word lines. Programming the memory cells of a word line to retain data typically occurs in a plurality of program loops, each of which includes the application of a programming pulse to a control gate of the word line and, optionally, a verify operation to sense the threshold voltages of the memory cells being programmed. Each program loop may also include a pre-charge operation prior to the programming pulse to pre-charge a plurality of channels containing memory cells to be programmed.

SUMMARY

One aspect of the present disclosure is related to a method of performing a programming operation in a memory device. The method includes the step of preparing a memory block that includes an array of memory cells that are arranged in a plurality of word lines. The word lines are in electrical communication with respective word line drivers and word line switches. In a program loop of a programming operation, the method continues with the step of applying an elevated voltage to a selected word line of the plurality of word lines. The method proceeds with the step of receiving a command to suspend the programming operation. With the word line switch associated with the selected word line turned on, the method continues with the step of ramping the selected word line from the elevated voltage to a reduced gate holding voltage and then turning off the word line switch associated with the selected word line to electrically isolate the selected word line from the associated word line driver so that the selected word line remains at the gate holding voltage until the programming operation resumes.

According to another aspect of the present disclosure, the elevated voltage is a programming voltage.

According to yet another aspect of the present disclosure, the elevated voltage is a reference voltage that is associated with one of a plurality of programmed data states that the memory cells of the selected word line are being programmed to.

According to still another aspect of the present disclosure, the gate holding voltage is in the range of two to eight Volts.

According to a further aspect of the present disclosure, after the step of turning the word line switch associated with the selected word line off, the method further includes the step of ramping down the word line driver associated with the selected word line from the gate holding voltage to approximately zero Volts while the selected word line remains at the gate holding voltage.

According to yet a further aspect of the present disclosure, during the programming operation, the memory cells of the selected word line are being programmed to at least three bits of data per memory cell.

According to still a further aspect of the present disclosure, the plurality of word lines further includes a plurality of unselected word lines. Following the step of receiving the instruction to suspend the programming operation, the method further includes the step of ramping the word line drivers that are associated with the unselected word lines directly to approximately zero Volts.

Another aspect of the present disclosure is related to a memory device. The memory device includes a memory block with an array of memory cells that are arranged in a plurality of word lines. The word lines are in electrical communication with respective word line drivers and word line switches. The memory device also includes control circuitry that is configured to program the memory cells of a selected word line of the plurality of word lines in a programming operation. During the programming operation, the control circuitry is configured to apply an elevated voltage to the selected word line and receive a command to suspend the programming operation. With the word line switch associated with the selected word line turned on, the control circuitry is configured to ramp the selected word line from the elevated voltage to a reduced gate holding voltage and then turn the word line switch associated with the selected word line off to electrically isolate the selected word line from the associated word line driver so that the selected word line remains at the gate holding voltage until the programming operation resumes.

According to another aspect of the present disclosure, the elevated voltage is a programming voltage.

According to yet another aspect of the present disclosure, the elevated voltage is a reference voltage that is associated with one of a plurality of programmed data states that the memory cells of the selected word line are being programmed to.

According to still another aspect of the present disclosure, the gate holding voltage is in the range of two to eight Volts.

According to a further aspect of the present disclosure, after turning off the word line switch associated with the selected word line, the method further includes the step of ramping down the word line driver associated with the selected word line from the gate holding voltage to approximately zero Volts while the selected word line remains at the gate holding voltage.

According to yet a further aspect of the present disclosure, during the programming operation, the control circuitry is programming the memory cells of the selected word line to at least three bits of data per memory cell.

According to still a further aspect of the present disclosure, the plurality of word lines further includes a plurality of unselected word lines, and following the control circuitry receiving the instruction to suspend the programming operation, the control circuitry ramps the word line drivers that are associated with the unselected word lines directly to approximately zero Volts.

Yet another aspect of the present disclosure is related to an apparatus that includes a memory block with an array of memory cells that are arranged in a plurality of word lines. The word lines are in electrical communication with respective word line drivers and word line switches. The apparatus also includes a programming means for programming the memory cells of a selected word line of the plurality of word lines in a programming operation. During the programming operation, the programming means is configured to apply an elevated voltage to the selected word line and receive a command to suspend the programming operation. With the word line switch associated with the selected word line turned on, the programming means ramps the selected word line from the elevated voltage to a reduced gate holding voltage and then turns the word line switch associated with the selected word line off to electrically isolate the selected word line from the associated word line driver so that the selected word line remains at the gate holding voltage until the programming operation resumes.

According to another aspect of the present disclosure, the elevated voltage is a programming voltage.

According to yet another aspect of the present disclosure, the elevated voltage is a reference voltage that is associated with one of a plurality of programmed data states that the memory cells of the selected word line are being programmed to.

According to still another aspect of the present disclosure, the gate holding voltage is in the range of two to eight Volts.

According to a further aspect of the present disclosure, after turning off the word line switch associated with the selected word line, the programming means ramps down the word line driver associated with the selected word line from the gate holding voltage to approximately zero Volts while the selected word line remains at the gate holding voltage.

According to yet a further aspect of the present disclosure, during the programming operation, the programming means is programming the memory cells of the selected word line to at least three bits of data per memory cell.

BRIEF DESCRIPTION OF THE DRAWINGS

A more detailed description is set forth below with reference to example embodiments depicted in the appended figures. Understanding that these figures depict only example embodiments of the disclosure and are, therefore, not to be considered limiting of its scope. The disclosure is described and explained with added specificity and detail through the use of the accompanying drawings in which:

FIG. **1**A is a block diagram of an example memory device;

FIG. **1**B is a block diagram of an example control circuit;

FIG. **1**C is a block diagram of example circuitry of the memory device of FIG. **1**A;

FIG. **2** depicts blocks of memory cells in an example two-dimensional configuration of the memory array of FIG. **1**A;

FIG. **3**A and FIG. **3**B depict cross-sectional views of example floating gate memory cells in NAND strings;

FIG. **4**A and FIG. **4**B depict cross-sectional views of example charge-trapping memory cells in NAND strings;

FIG. **5** depicts an example block diagram of the sense block SB**1** of FIG. **1**;

FIG. **6**A is a perspective view of a set of blocks in an example three-dimensional configuration of the memory array of FIG. **1**;

FIG. **6**B depicts an example cross-sectional view of a portion of one of the blocks of FIG. **6**A;

FIG. **6**C depicts a plot of memory hole diameter in the stack of FIG. **6**B;

FIG. **6**D depicts a close-up view of region **622** of the stack of FIG. **6**B;

FIG. **7**A depicts a top view of an example word line layer WL**0** of the stack of FIG. **6**B;

FIG. **7**B depicts a top view of an example top dielectric layer DL**116** of the stack of FIG. **6**B;

FIG. **8** depicts a threshold voltage distribution of a page of memory cells programmed to one bit per memory cell (SLC);

FIG. **9** depicts a threshold voltage distribution of a page of memory cells programmed to three bits per memory cell (TLC);

FIG. **10** is a waveform of the voltages applied to a selected word line during an example programming operation;

FIG. **11** is a time chart illustrating the steps performed during an example programming operation;

FIG. **12** is a plot depicting the threshold voltage of an example memory cell during a programming operation in which a suspension event occurs;

FIG. **13** is another plot depicting the threshold voltage of an example memory cell during a programming operation in which a different suspension event occurs;

FIG. **14** is a schematic view of an example memory cell during a suspension event;

FIG. **15** is a schematic view of an example NAND string and the associated word line drivers and word line switches;

FIG. **16** is a plot of the voltages applied to a selected word line driver (WL Driver), a selected word line switch (WLSW) and a selected word line WLn during a suspend-resume-go operation according to an exemplary embodiment of the present disclosure; and

FIG. **17** is a flow chart depicting the steps of performing a programming operation during which a suspension event occurs and a suspend-resume-go operation according to an exemplary embodiment of the present disclosure is employed.

DESCRIPTION OF THE ENABLING EMBODIMENTS

According to an aspect of the subject disclosure, a programming technique is provided for avoiding under-programming and over-programming in the event that a suspension event occurs during a programming operation in a selected word line WLn. More specifically, when an instruction to suspend the programming operation is received, the voltage that is applied to a word line driver that is associated with the selected word line WLn is ramped to a control gate hold voltage. Next, a word line switch associated with the selected word line WLn is turned off (mode non-conductive to electrons). This isolates the selected word line WLn from the associated word line driver. The word line driver can then ramp down to approximately zero Volts (0 V) and the selected word line WLn will remain at the control gate hold voltage throughout the suspension event. By maintaining the selected word line WLn at the control gate hold voltage, short term data retention (STDR), which normally occurs during suspension events, is reduced, thereby preventing under-programming and over-programming. These steps and others are discussed in further detail below.

FIG. **1**A is a block diagram of an example memory device **100** is configured to program the memory cells in the word lines of a memory block according to the programming techniques of the subject disclosure. The memory die **108** includes a memory structure **126** of memory cells, such as an array of memory cells, control circuitry **110**, and read/write circuits **128**. The memory structure **126** is addressable by word lines via a row decoder **124** and by bit lines via a column decoder **132**. The read/write circuits **128** include multiple sense blocks SB**1**, SB**2**, . . . SBp (sensing circuitry) and allow a page of memory cells to be read or programmed in parallel. Typically, a controller **122** is included in the same memory device **100** (e.g., a removable storage card) as the one or more memory die **108**. Commands and data are transferred between the host **140** and controller **122** via a data bus **120**, and between the controller and the one or more memory die **108** via lines **118**.

The memory structure **126** can be two-dimensional or three-dimensional. The memory structure **126** may comprise one or more array of memory cells including a three-dimensional array. The memory structure **126** may comprise a monolithic three-dimensional memory structure in which multiple memory levels are formed above (and not in) a single substrate, such as a wafer, with no intervening substrates. The memory structure **126** may comprise any type of non-volatile memory that is monolithically formed in one or more physical levels of arrays of memory cells having an active area disposed above a silicon substrate. The memory structure **126** may be in a non-volatile memory device having circuitry associated with the operation of the memory cells, whether the associated circuitry is above or within the substrate.

The control circuitry **110** cooperates with the read/write circuits **128** to perform memory operations on the memory structure **126**, and includes a state machine **112**, an on-chip address decoder **114**, and a power control module **116**. The state machine **112** provides chip-level control of memory operations.

A storage region **113** may, for example, be provided for programming parameters. The programming parameters may include a program voltage, a program voltage bias, position parameters indicating positions of memory cells, contact line connector thickness parameters, a verify voltage, and/or the like. The position parameters may indicate a position of a memory cell within the entire array of NAND strings, a position of a memory cell as being within a particular NAND string group, a position of a memory cell on a particular plane, and/or the like. The contact line connector thickness parameters may indicate a thickness of a contact line connector, a substrate or material that the contact line connector is comprised of, and/or the like.

The on-chip address decoder **114** provides an address interface between that used by the host or a memory controller to the hardware address used by the decoders **124** and **132**. The power control module **116** controls the power and voltages supplied to the word lines and bit lines during memory operations. It can include drivers for word lines, SGS and SGD transistors, and source lines. The sense blocks can include bit line drivers, in one approach. An SGS transistor is a select gate transistor at a source end of a NAND string, and an SGD transistor is a select gate transistor at a drain end of a NAND string.

In some embodiments, some of the components can be combined. In various designs, one or more of the components (alone or in combination), other than memory structure **126**, can be thought of as at least one control circuit which is configured to perform the actions described herein. For example, a control circuit may include any one of, or a combination of, control circuitry **110**, state machine **112**, decoders **114**/**132**, power control module **116**, sense blocks SBb, SB**2**, . . . , SBp, read/write circuits **128**, controller **122**, and so forth.

The control circuits **150** can include a programming circuit **151** configured to perform a program and verify operation for one set of memory cells, wherein the one set of memory cells comprises memory cells assigned to represent one data state among a plurality of data states and memory cells assigned to represent another data state among the plurality of data states; the program and verify operation comprising a plurality of program and verify iterations; and in each program and verify iteration, the programming circuit performs programming for the one selected word line after which the programming circuit applies a verification signal to the selected word line. The control circuits **150** can also include a counting circuit **152** configured to obtain a count of memory cells which pass a verify test for the one data state. The control circuits **150** can also include a determination circuit **153** configured to determine, based on an amount by which the count exceeds a threshold, if a programming operation is completed.

For example, FIG. **1**B is a block diagram of an example control circuit **150** which comprises the programming circuit **151**, the counting circuit **152**, and the determination circuit **153**.

The off-chip controller **122** may comprise a processor **122***c*, storage devices (memory) such as ROM **122***a* and RAM **122***b*, an error-correction code (ECC) engine **245**, and a suspend-resume-go (SRG) engine **246**. The ECC engine can correct a number of read errors which are caused when the upper tail of a Vt distribution becomes too high. However, uncorrectable errors may exist in some cases. The techniques provided herein reduce the likelihood of uncorrectable errors. The SRG engine **246** can perform the programming techniques discussed in further detail below.

The storage device(s) **122***a*, **122***b* comprise, code such as a set of instructions, and the processor **122***c* is operable to execute the set of instructions to provide the functionality described herein. Alternately or additionally, the processor **122***c* can access code from a storage device **126***a* of the memory structure **126**, such as a reserved area of memory cells in one or more word lines. For example, code can be used by the controller **122** to access the memory structure **126** such as for programming, read and erase operations. The code can include boot code and control code (e.g., set of instructions). The boot code is software that initializes the controller **122** during a booting or startup process and enables the controller **122** to access the memory structure **126**. The code can be used by the controller **122** to control one or more memory structures **126**. Upon being powered up, the processor **122***c* fetches the boot code from the ROM **122***a* or storage device **126***a* for execution, and the boot code initializes the system components and loads the control code into the RAM **122***b*. Once the control code is loaded into the RAM **122***b*, it is executed by the processor **122***c*. The control code includes drivers to perform basic tasks such as controlling and allocating memory, prioritizing the processing of instructions, and controlling input and output ports.

Generally, the control code can include instructions to perform the functions described herein including the steps of the flowcharts discussed further below and provide the voltage waveforms including those discussed further below. For example, as illustrated in FIG. **1**C, the control circuitry **110**, controller **122**, control circuits **150**, and/or any other circuitry are configured/programmed, during a programming operation, at step **160** to receive an instruction to begin a suspension event. The suspension event could be, for example, the memory device performing a read operation in another memory block. At step **161**, a voltage that is applied to a word line driver (WL driver) that is associated with a selected word line WLn being programmed is reduced from an elevated voltage, such as a programming voltage or a reference voltage, to a control gate hold voltage PSRG. At step **162**, the word line switch (WLSW) that is associated with the selected word sline WLn is turned off (made non-conductive to electrons). By turning off the WLSW, the selected word line WLn is electrically isolated from the WL driver such that the selected word line WLn remains at the control gate hold voltage PSRG. At step **163**, the WL driver associated with the selected word line WLn is reduced to approximately zero Volts (0 V).

In one embodiment, the host is a computing device (e.g., laptop, desktop, smartphone, tablet, digital camera) that includes one or more processors, one or more processor readable storage devices (RAM, ROM, flash memory, hard disk drive, solid state memory) that store processor readable code (e.g., software) for programming the one or more processors to perform the methods described herein. The host may also include additional system memory, one or more input/output interfaces and/or one or more input/output devices in communication with the one or more processors.

Other types of non-volatile memory in addition to NAND flash memory can also be used.

Semiconductor memory devices include volatile memory devices, such as dynamic random access memory ("DRAM") or static random access memory ("SRAM") devices, non-volatile memory devices, such as resistive random access memory ("ReRAM"), electrically erasable programmable read only memory ("EEPROM"), flash memory (which can also be considered a subset of EEPROM), ferroelectric random access memory ("FRAM"), and magnetoresistive random access memory ("MRAM"), and other semiconductor elements capable of storing information. Each type of memory device may have different configurations. For example, flash memory devices may be configured in a NAND or a NOR configuration.

The memory devices can be formed from passive and/or active elements, in any combinations. By way of non-limiting example, passive semiconductor memory elements include ReRAM device elements, which in some embodiments include a resistivity switching storage element, such as an anti-fuse or phase change material, and optionally a steering element, such as a diode or transistor. Further by way of non-limiting example, active semiconductor memory elements include EEPROM and flash memory device elements, which in some embodiments include elements containing a charge storage region, such as a floating gate, conductive nanoparticles, or a charge storage dielectric material.

Multiple memory elements may be configured so that they are connected in series or so that each element is individually accessible. By way of non-limiting example, flash memory devices in a NAND configuration (NAND memory) typically contain memory elements connected in series. A NAND string is an example of a set of series-connected transistors comprising memory cells and SG transistors.

A NAND memory array may be configured so that the array is composed of multiple memory strings in which a string is composed of multiple memory elements sharing a single bit line and accessed as a group. Alternatively, memory elements may be configured so that each element is individually accessible, e.g., a NOR memory array. NAND and NOR memory configurations are examples, and memory elements may be otherwise configured. The semiconductor memory elements located within and/or over a substrate may be arranged in two or three dimensions, such as a two-dimensional memory structure or a three-dimensional memory structure.

In a two-dimensional memory structure, the semiconductor memory elements are arranged in a single plane or a single memory device level. Typically, in a two-dimensional memory structure, memory elements are arranged in a plane (e.g., in an x-y direction plane) which extends substantially parallel to a major surface of a substrate that supports the memory elements. The substrate may be a wafer over or in which the layer of the memory elements is formed or it may be a carrier substrate which is attached to the memory elements after they are formed. As a non-limiting example, the substrate may include a semiconductor such as silicon.

The memory elements may be arranged in the single memory device level in an ordered array, such as in a plurality of rows and/or columns. However, the memory elements may be arrayed in non-regular or non-orthogonal configurations. The memory elements may each have two or more electrodes or contact lines, such as bit lines and word lines.

A three-dimensional memory array is arranged so that memory elements occupy multiple planes or multiple memory device levels, thereby forming a structure in three dimensions (i.e., in the x, y and z directions, where the z-direction is substantially perpendicular and the x- and y-directions are substantially parallel to the major surface of the substrate).

As a non-limiting example, a three-dimensional memory structure may be vertically arranged as a stack of multiple two-dimensional memory device levels. As another non-limiting example, a three-dimensional memory array may be arranged as multiple vertical columns (e.g., columns extending substantially perpendicular to the major surface of the substrate, i.e., in the y direction) with each column having multiple memory elements. The columns may be arranged in a two-dimensional configuration, e.g., in an x-y plane, resulting in a three-dimensional arrangement of memory elements with elements on multiple vertically stacked memory planes. Other configurations of memory elements in three dimensions can also constitute a three-dimensional memory array.

By way of non-limiting example, in a three-dimensional array of NAND strings, the memory elements may be coupled together to form a NAND string within a single horizontal (e.g., x-y) memory device level. Alternatively, the memory elements may be coupled together to form a vertical NAND string that traverses across multiple horizontal memory device levels. Other three-dimensional configurations can be envisioned wherein some NAND strings contain memory elements in a single memory level while other strings contain memory elements which span through multiple memory levels. Three-dimensional memory arrays may also be designed in a NOR configuration and in a ReRAM configuration.

Typically, in a monolithic three-dimensional memory array, one or more memory device levels are formed above a single substrate. Optionally, the monolithic three-dimensional memory array may also have one or more memory layers at least partially within the single substrate. As a non-limiting example, the substrate may include a semiconductor such as silicon. In a monolithic three-dimensional array, the layers constituting each memory device level of the array are typically formed on the layers of the underlying memory device levels of the array. However, layers of adjacent memory device levels of a monolithic three-dimensional memory array may be shared or have intervening layers between memory device levels.

Then again, two-dimensional arrays may be formed separately and then packaged together to form a non-monolithic memory device having multiple layers of memory. For example, non-monolithic stacked memories can be constructed by forming memory levels on separate substrates and then stacking the memory levels atop each other. The substrates may be thinned or removed from the memory device levels before stacking, but as the memory device levels are initially formed over separate substrates, the resulting memory arrays are not monolithic three-dimensional memory arrays. Further, multiple two-dimensional memory arrays or three-dimensional memory arrays (monolithic or non-monolithic) may be formed on separate chips and then packaged together to form a stacked-chip memory device.

FIG. **2** illustrates memory blocks **200**, **210** of memory cells in an example two-dimensional configuration of the memory array **126** of FIG. **1**. The memory array **126** can include many such blocks **200**, **210**. Each example block **200**, **210** includes a number of NAND strings and respective bit lines, e.g., BL**0**, BL**1**, . . . which are shared among the blocks. Each NAND string is connected at one end to a drain-side select gate (SGD), and the control gates of the drain-side select gates are connected via a common SGD line. The NAND strings are connected at their other end to a source-side select gate (SGS) which, in turn, is connected to a common source line **220**. One hundred and twelve word lines, for example, WL**0**-WL**111**, extend between the SGSs and the SGDs. In some embodiments, the memory block may include more or fewer than one hundred and twelve word lines. For example, in some embodiments, a memory block includes one hundred and sixty-four word lines. In some cases, dummy word lines, which contain no user data, can also be used in the memory array adjacent to the select gate transistors or between certain data word lines. Such dummy word lines can shield the edge data word line from certain edge effects.

One type of non-volatile memory which may be provided in the memory array is a floating gate memory, such as of the type shown in FIGS. **3**A and **3**B. However, other types of non-volatile memory can also be used. As discussed in further detail below, in another example shown in FIGS. **4**A and **4**B, a charge-trapping memory cell uses a non-conductive dielectric material in place of a conductive floating gate to store charge in a non-volatile manner. A triple layer dielectric formed of silicon oxide, silicon nitride and silicon oxide ("ONO") is sandwiched between a conductive control gate and a surface of a semi-conductive substrate above the memory cell channel. The cell is programmed by injecting electrons from the cell channel into the nitride, where they are trapped and stored in a limited region. This stored charge then changes the threshold voltage of a portion of the channel of the cell in a manner that is detectable. The cell is erased by injecting hot holes into the nitride. A similar cell can be provided in a split-gate configuration where a doped polysilicon gate extends over a portion of the memory cell channel to form a separate select transistor.

In another approach, NROM cells are used. Two bits, for example, are stored in each NROM cell, where an ONO dielectric layer extends across the channel between source and drain diffusions. The charge for one data bit is localized in the dielectric layer adjacent to the drain, and the charge for the other data bit localized in the dielectric layer adjacent to the source. Multi-state data storage is obtained by separately reading binary states of the spatially separated charge storage regions within the dielectric. Other types of non-volatile memory are also known.

FIG. **3**A illustrates a cross-sectional view of example floating gate memory cells **300**, **310**, **320** in NAND strings. In this Figure, a bit line or NAND string direction goes into the page, and a word line direction goes from left to right. As an example, word line **324** extends across NAND strings which include respective channel regions **306**, **316** and **326**. The memory cell **300** includes a control gate **302**, a floating gate **304**, a tunnel oxide layer **305** and the channel region **306**. The memory cell **310** includes a control gate **312**, a floating gate **314**, a tunnel oxide layer **315** and the channel region **316**. The memory cell **320** includes a control gate **322**, a floating gate **321**, a tunnel oxide layer **325** and the channel region **326**. Each memory cell **300**, **310**, **320** is in a different respective NAND string. An inter-poly dielectric (IPD) layer **328** is also illustrated. The control gates **302**, **312**, **322** are portions of the word line. A cross-sectional view along contact line connector **329** is provided in FIG. **3**B.

The control gate **302**, **312**, **322** wraps around the floating gate **304**, **314**, **321**, increasing the surface contact area between the control gate **302**, **312**, **322** and floating gate **304**, **314**, **321**. This results in higher IPD capacitance, leading to a higher coupling ratio which makes programming and erase easier. However, as NAND memory devices are scaled down, the spacing between neighboring cells **300**, **310**, **320** becomes smaller so there is almost no space for the control gate **302**, **312**, **322** and the IPD layer **328** between two adjacent floating gates **302**, **312**, **322**.

As an alternative, as shown in FIGS. **4**A and **4**B, the flat or planar memory cell **400**, **410**, **420** has been developed in which the control gate **402**, **412**, **422** is flat or planar; that is, it does not wrap around the floating gate and its only contact with the charge storage layer **428** is from above it. In this case, there is no advantage in having a tall floating gate. Instead, the floating gate is made much thinner. Further, the floating gate can be used to store charge, or a thin charge trap layer can be used to trap charge. This approach can avoid the issue of ballistic electron transport, where an electron can travel through the floating gate after tunneling through the tunnel oxide during programming.

FIG. **4**A depicts a cross-sectional view of example charge-trapping memory cells **400**, **410**, **420** in NAND strings. The view is in a word line direction of memory cells **400**, **410**, **420** comprising a flat control gate and charge-trapping regions as a two-dimensional example of memory cells **400**, **410**, **420** in the memory cell array **126** of FIG. **1**. Charge-trapping memory can be used in NOR and NAND flash memory device. This technology uses an insulator such as an SiN film to store electrons, in contrast to a floating-gate MOSFET technology which uses a conductor such as doped polycrystalline silicon to store electrons. As an example, a word line **424** extends across NAND strings which include respective channel regions **406**, **416**, **426**. Portions of the word line provide control gates **402**, **412**, **422**. Below the word line is an IPD layer **428**, charge-trapping layers **404**, **414**, **421**, polysilicon layers **405**, **415**, **425**, and tunneling layers **409**, **407**, **408**. Each charge-trapping layer **404**, **414**, **421** extends continuously in a respective NAND string. The flat configuration of the control gate can be made thinner than a floating gate. Additionally, the memory cells can be placed closer together.

FIG. **4**B illustrates a cross-sectional view of the structure of FIG. **4**A along contact line connector **429**. The NAND string **430** includes an SGS transistor **431**, example memory cells **400**, **433**, . . . **435**, and an SGD transistor **436**. Passageways in the IPD layer **428** in the SGS and SGD transistors **431**, **436** allow the control gate layers **402** and floating gate layers to communicate. The control gate **402** and floating gate layers may be polysilicon and the tunnel oxide layer may be silicon oxide, for instance. The IPD layer

428 can be a stack of nitrides (N) and oxides (O) such as in a N—O—N—O—N configuration.

The NAND string may be formed on a substrate which comprises a p-type substrate region 455, an n-type well 456 and a p-type well 457. N-type source/drain diffusion regions sd1, sd2, sd3, sd4, sd5, sd6 and sd7 are formed in the p-type well. A channel voltage, Vch, may be applied directly to the channel region of the substrate.

FIG. 5 illustrates an example block diagram of the sense block SB1 of FIG. 1. In one approach, a sense block comprises multiple sense circuits. Each sense circuit is associated with data latches. For example, the example sense circuits 550*a*, 551*a*, 552*a*, and 553*a* are associated with the data latches 550*b*, 551*b*, 552*b*, and 553*b*, respectively. In one approach, different subsets of bit lines can be sensed using different respective sense blocks. This allows the processing load which is associated with the sense circuits to be divided up and handled by a respective processor in each sense block. For example, a sense circuit controller 560 in SB1 can communicate with the set of sense circuits and latches. The sense circuit controller 560 may include a pre-charge circuit 561 which provides a voltage to each sense circuit for setting a pre-charge voltage. In one possible approach, the voltage is provided to each sense circuit independently, e.g., via the data bus and a local bus. In another possible approach, a common voltage is provided to each sense circuit concurrently. The sense circuit controller 560 may also include a pre-charge circuit 561, a memory 562 and a processor 563. The memory 562 may store code which is executable by the processor to perform the functions described herein. These functions can include reading the latches 550*b*, 551*b*, 552*b*, 553*b* which are associated with the sense circuits 550*a*, 551*a*, 552*a*, 553*a*, setting bit values in the latches and providing voltages for setting pre-charge levels in sense nodes of the sense circuits 550*a*, 551*a*, 552*a*, 553*a*. Further example details of the sense circuit controller 560 and the sense circuits 550*a*, 551*a*, 552*a*, 553*a* are provided below.

In some embodiments, a memory cell may include a flag register that includes a set of latches storing flag bits. In some embodiments, a quantity of flag registers may correspond to a quantity of data states. In some embodiments, one or more flag registers may be used to control a type of verification technique used when verifying memory cells. In some embodiments, a flag bit's output may modify associated logic of the device, e.g., address decoding circuitry, such that a specified block of cells is selected. A bulk operation (e.g., an erase operation, etc.) may be carried out using the flags set in the flag register, or a combination of the flag register with the address register, as in implied addressing, or alternatively by straight addressing with the address register alone.

FIG. 6A is a perspective view of a set of blocks 600 in an example three-dimensional configuration of the memory array 126 of FIG. 1. On the substrate are example blocks BLK0, BLK1, BLK2, BLK3 of memory cells (storage elements) and a peripheral area 604 with circuitry for use by the blocks BLK0, BLK1, BLK2, BLK3. For example, the circuitry can include voltage drivers 605 which can be connected to control gate layers of the blocks BLK0, BLK1, BLK2, BLK3. In one approach, control gate layers at a common height in the blocks BLK0, BLK1, BLK2, BLK3 are commonly driven. The substrate 601 can also carry circuitry under the blocks BLK0, BLK1, BLK2, BLK3, along with one or more lower metal layers which are patterned in conductive paths to carry signals of the circuitry. The blocks BLK0, BLK1, BLK2, BLK3 are formed in an intermediate region 602 of the memory device. In an upper region 603 of the memory device, one or more upper metal layers are patterned in conductive paths to carry signals of the circuitry. Each block BLK0, BLK1, BLK2, BLK3 comprises a stacked area of memory cells, where alternating levels of the stack represent word lines. In one possible approach, each block BLK0, BLK1, BLK2, BLK3 has opposing tiered sides from which vertical contacts extend upward to an upper metal layer to form connections to conductive paths. While four blocks BLK0, BLK1, BLK2, BLK3 are illustrated as an example, two or more blocks can be used, extending in the x- and/or y-directions.

In one possible approach, the length of the plane, in the x-direction, represents a direction in which signal paths to word lines extend in the one or more upper metal layers (a word line or SGD line direction), and the width of the plane, in the y-direction, represents a direction in which signal paths to bit lines extend in the one or more upper metal layers (a bit line direction). The z-direction represents a height of the memory device.

FIG. 6B illustrates an example cross-sectional view of a portion of one of the blocks BLK0, BLK1, BLK2, BLK3 of FIG. 6A. The block comprises a stack 610 of alternating conductive and dielectric layers. In this example, the conductive layers comprise two SGD layers, two SGS layers and four dummy word line layers DWLD0, DWLD1, DWLS0 and DWLS1, in addition to data word line layers (word lines) WL0-WL111. The dielectric layers are labelled as DL0-DL116. Further, regions of the stack 610 which comprise NAND strings NS1 and NS2 are illustrated. Each NAND string encompasses a memory hole 618, 619 which is filled with materials which form memory cells adjacent to the word lines. A region 622 of the stack 610 is shown in greater detail in FIG. 6D and is discussed in further detail below. The dielectric layers can have variable thicknesses such that some of the conductive layers can be closer to or further from neighboring conductive layers. The thicknesses of the dielectric layers affects the "ON pitch," which is a factor in memory density. Specifically, a smaller ON pitch allows for more memory cells in a given area but may compromise reliability.

The stack 610 includes a substrate 611, an insulating film 612 on the substrate 611, and a portion of a source line SL. NS1 has a source-end 613 at a bottom 614 of the stack and a drain-end 615 at a top 616 of the stack 610. Contact line connectors (e.g., slits, such as metal-filled slits) 617, 620 may be provided periodically across the stack 610 as interconnects which extend through the stack 610, such as to connect the source line to a particular contact line above the stack 610. The contact line connectors 617, 620 may be used during the formation of the word lines and subsequently filled with metal. A portion of a bit line BL0 is also illustrated. A conductive via 621 connects the drain-end 615 to BL0.

FIG. 6C illustrates a plot of memory hole diameter in the stack of FIG. 6B. The vertical axis is aligned with the stack of FIG. 6B and illustrates a width (wMH), e.g., diameter, of the memory holes 618 and 619. The word line layers WL0-WL111 of FIG. 6A are repeated as an example and are at respective heights z0-z111 in the stack. In such a memory device, the memory holes which are etched through the stack have a very high aspect ratio. For example, a depth-to-diameter ratio of about 25-30 is common. The memory holes may have a circular cross-section. Due to the etching process, the memory hole width can vary along the length of the hole. Typically, the diameter becomes progressively smaller from the top to the bottom of the memory hole. That is, the memory holes are tapered, narrowing at the bottom of the stack. In some cases, a slight narrowing occurs at the top of the hole near the select gate so that the diameter becomes slightly wider before becoming progressively smaller from the top to the bottom of the memory hole.

FIG. **6**D illustrates a close-up view of the region **622** of the stack **610** of FIG. **6**B. Memory cells are formed at the different levels of the stack at the intersection of a word line layer and a memory hole. In this example, SGD transistors **680**, **681** are provided above dummy memory cells **682**, **683** and a data memory cell MC. A number of layers can be deposited along the sidewall (SW) of the memory hole **630** and/or within each word line layer, e.g., using atomic layer deposition. For example, each column (e.g., the pillar which is formed by the materials within a memory hole **630**) can include a charge-trapping layer or film **663** such as SiN or other nitride, a tunneling layer **664**, a polysilicon body or channel **665**, and a dielectric core **666**. A word line layer can include a blocking oxide/block high-k material **660**, a metal barrier **661**, and a conductive metal such as Tungsten as a control gate. For example, control gates **690**, **691**, **692**, **693**, and **694** are provided. In this example, all of the layers except the metal are provided in the memory hole **630**. In other approaches, some of the layers can be in the control gate layer. Additional pillars are similarly formed in the different memory holes. A pillar can form a columnar active area (AA) of a NAND string.

When a memory cell is programmed, electrons are stored in a portion of the charge-trapping layer which is associated with the memory cell. These electrons are drawn into the charge-trapping layer from the channel, and through the tunneling layer. The threshold voltage Vt of a memory cell is increased in proportion to the amount of stored charge. During a sensing operation, the threshold voltage Vt is detected or measured. During an erase operation, the electrons return to the channel.

Each of the memory holes **630** can be filled with a plurality of annular layers comprising a blocking oxide layer, a charge trapping layer **663**, a tunneling layer **664** and a channel layer. A core region of each of the memory holes **630** is filled with a body material, and the plurality of layers are between the core region and the word line layer in each of the memory holes **630**. In some cases, the charge trapping layer **663** and the tunneling layer **664** are annular in shape. In other cases, as discussed in further detail below, these layers are semi-circular in shape.

The NAND string can be considered to have a floating body channel because the length of the channel is not formed on a substrate. Further, the NAND string is provided by a plurality of word line layers above one another in a stack, and separated from one another by dielectric layers.

FIG. **7**A illustrates a top view of an example word line layer WL**0** of the stack **610** of FIG. **6**B. As mentioned, a three-dimensional memory device can comprise a stack of alternating conductive and dielectric layers. The conductive layers provide the control gates of the SG transistors and memory cells. The layers used for the SG transistors are SG layers and the layers used for the memory cells are word line layers. Further, memory holes are formed in the stack and filled with a charge-trapping material and a channel material. As a result, a vertical NAND string is formed. Source lines are connected to the NAND strings below the stack and bit lines are connected to the NAND strings above the stack.

A block BLK in a three-dimensional memory device can be divided into sub-blocks, where each sub-block comprises a NAND string group which has a common SGD control line. For example, see the SGD lines/control gates SGD**0**, SGD**1**, SGD**2** and SGD**3** in the sub-blocks SBa, SBb, SBc and SBd, respectively. Further, a word line layer in a block can be divided into regions. Each region is in a respective sub-block and can extend between contact line connectors (e.g., slits) which are formed periodically in the stack to process the word line layers during the fabrication process of the memory device. This processing can include replacing a sacrificial material of the word line layers with metal. Generally, the distance between contact line connectors should be relatively small to account for a limit in the distance that an etchant can travel laterally to remove the sacrificial material, and that the metal can travel to fill a void which is created by the removal of the sacrificial material. For example, the distance between contact line connectors may allow for a few rows of memory holes between adjacent contact line connectors. The layout of the memory holes and contact line connectors should also account for a limit in the number of bit lines which can extend across the region while each bit line is connected to a different memory cell. After processing the word line layers, the contact line connectors can optionally be filed with metal to provide an interconnect through the stack.

In this example, there are four rows of memory holes between adjacent contact line connectors. A row here is a group of memory holes which are aligned in the x-direction. Moreover, the rows of memory holes are in a staggered pattern to increase the density of the memory holes. The word line layer or word line is divided into regions WL**0***a*, WL**0***b*, WL**0***c* and WL**0***d* which are each connected by a contact line **713**. The last region of a word line layer in a block can be connected to a first region of a word line layer in a next block, in one approach. The contact line **713**, in turn, is connected to a voltage driver for the word line layer. The region WL**0***a* has example memory holes **710**, **711** along a contact line **712**. The region WL**0***b* has example memory holes **714**, **715**. The region WL**0***c* has example memory holes **716**, **717**. The region WL**0***d* has example memory holes **718**, **719**. The memory holes are also shown in FIG. **7**B. Each memory hole can be part of a respective NAND string. For example, the memory holes **710**, **714**, **716** and **718** can be part of NAND strings NS**0**_SBa, NS**1**_SBb, NS**2**_SBc, NS**3**_SBd, and NS**4**_SBe, respectively.

Each circle represents the cross-section of a memory hole at a word line layer or SG layer. Example circles shown with dashed lines represent memory cells which are provided by the materials in the memory hole and by the adjacent word line layer. For example, memory cells **720**, **721** are in WL**0***a*, memory cells **724**, **725** are in WL**0***b*, memory cells **726**, **727** are in WL**0***c*, and memory cells **728**, **729** are in WL**0***d*. These memory cells are at a common height in the stack.

Contact line connectors (e.g., slits, such as metal-filled slits) **701**, **702**, **703**, **704** may be located between and adjacent to the edges of the regions WL**0***a*-WL**0***d*. The contact line connectors **701**, **702**, **703**, **704** provide a conductive path from the bottom of the stack to the top of the stack. For example, a source line at the bottom of the stack may be connected to a conductive line above the stack, where the conductive line is connected to a voltage driver in a peripheral region of the memory device.

FIG. **7**B illustrates a top view of an example top dielectric layer DL**116** of the stack of FIG. **6**B. The dielectric layer is divided into regions DL**116***a*, DL**116***b*, DL**116***c* and DL**116***d*. Each region can be connected to a respective voltage driver. This allows a set of memory cells in one region of a word line layer being programmed concurrently, with each memory cell being in a respective NAND string which is connected to a respective bit line. A voltage can be set on each bit line during each programming, sensing, or erasing operation.

The region DL**116***a* has the example memory holes **710**, **711** along a contact line **712**, which is coincident with a bit line BL**0**. A number of bit lines extend above the memory holes and are connected to the memory holes as indicated by the "X" symbols. BL**0** is connected to a set of memory holes which includes the memory holes **711**, **715**, **717**, **719**. Another example bit line BL**1** is connected to a set of memory holes which includes the memory holes **710**, **714**, **716**, **718**. The contact line connectors (e.g., slits, such as metal-filled slits) **701**, **702**, **703**, **704** from FIG. **7**A are also illustrated, as they extend vertically through the stack. The bit lines can be numbered in a sequence BL**0**-BL**23** across the DL**116** layer in the x-direction.

Different subsets of bit lines are connected to memory cells in different rows. For example, BL**0**, BL**4**, BL**8**, BL**12**, BL**16**, BL**20** are connected to memory cells in a first row of cells at the right-hand edge of each region. BL**2**, BL**6**, BL**10**, BL**14**, BL**18**, BL**22** are connected to memory cells in an adjacent row of cells, adjacent to the first row at the right-hand edge. BL**3**, BL**7**, BL**11**, BL**15**, BL**19**, BL**23** are connected to memory cells in a first row of cells at the left-hand edge of each region. BL**1**, BL**5**, BL**9**, BL**13**, BL**17**, BL**21** are connected to memory cells in an adjacent row of memory cells, adjacent to the first row at the left-hand edge.

The memory cells of the memory blocks can be programmed to store one or more bits of data in multiple data states, each of which is associated with a respective threshold voltage Vt range and with a respective bit or series of bits. For example, FIG. **8** depicts a threshold voltage Vt distribution of a group of memory cells programmed according to a one bit per memory cell (SLC) storage scheme. In the SLC storage scheme, there are two total data states, including the erased state (Er) and a single programmed data state (S1). FIG. **9** illustrates the threshold voltage Vt distribution of a three bits per cell (TLC) storage scheme that includes eight total data states, namely the erased state (Er) and seven programmed data states (S1, S2, S3, S4, S5, S6, and S7). Each programmed data state (S1-S7) is associated with a respective verify voltage (Vv**1**-Vv**7**), which is employed during a verify portion of a programming operation. Similarly, each programmed data state is associated with a unique read voltage that can be the same or different than the respective verify voltages. Other storage schemes are also available, such as two bits per cell (MLC) with four data states, four bits per cell (QLC) with sixteen data states, or five bits per cell (PLC) with thirty-two data states.

Programming the memory cells occurs on a word line-by-word line basis from one side of the memory block towards an opposite side of the memory block. In contrast, erase typically occurs on a block or sub-block basis. Typically, programming the memory cells of a selected word line to retain multiple bits per memory cell (for example, MLC, TLC, or QLC) starts with the memory cells being in the erased data state and includes a plurality of program loops. Each program loop includes both a programming pulse and a verify operation. FIG. **10** depicts a waveform **1000** of the voltages applied to a selected word line during an example programming operation for programming the memory cells of the selected word line to a greater number of bits per memory cell (e.g., TLC or QLC). As depicted, each program loop includes a programming pulse (hereinafter referred to as a VPGM pulse) and one or more verify pulses, depending on which data states are being programmed in a particular program loop. A square waveform is depicted for each pulse for simplicity; however, other shapes are possible, such as a multilevel shape or a ramped shape.

Incremental Step Pulse Programming (ISPP) is used in this example pulse train, which means that the VPGM pulse voltage steps up, or increases, in each successive program loop. More specifically, the pulse train includes VPGM pulses that increase stepwise in amplitude with each successive program loop by a program voltage step size (dVPGM). As discussed in further detail below, following a suspension event, the magnitude of the voltage step size could vary from a baseline. A new pulse train starts with the VPGM pulse being at a starting voltage VPGMU and ends with it being at a final VPGM pulse, which does not exceed a maximum allowed voltage. The example pulse train **1000** includes a series of VPGM pulses **1001**-**1018** that are applied to a control gate of the selected word line to program the memory cells of that word line and that increase in amplitude by the program voltage step size dVPGM between pulses.

One or more verify pulses **1020**-**1036** are provided after each VPGM pulse, based on the target data states which are being verified in the respective program loops. The verify voltages may be the voltages Vv**1**-Vv**7** shown in FIG. **9**. Concurrent with the application of the verify voltages, a sensing operation can determine whether a particular memory cell in the selected word line has a threshold voltage Vt above the a reference voltage VCG, which may be a verify voltage Vv associated with an intended data state for the particular selected memory cell. More specifically, a sense node is discharged through a NAND string containing the selected memory cell. If the current through the NAND string is relatively high, then the selected memory cell has a threshold voltage Vt that is below the reference voltage VCG and verify fails. On the other hand, if the current through the NAND string is relatively low, then the selected memory cell has a threshold voltage Vt that is above the reference voltage VCG and verify passes.

If the memory cell passes verify, programming of that memory cell is completed and further programming of that memory cell is inhibited (or locked out) for all remaining program loops by applying an inhibit voltage to a bit line coupled with the memory cell concurrent with the VPGM pulse and by skipping verify for those memory cells. Programming proceeds until all (or a sufficient number of) memory cells of the selected word line WLn pass verify for their intended states, in which case, programming passes, or until a predetermined maximum number of program loops is exceeded, in which case, programming fails.

Programming of a memory cell can be slowed when the memory cell has a threshold voltage Vt which is close to but below the threshold voltage Vt range of its intended data state. Specifically, in a Quick Pass Write (QPW) technique, each programmed data state includes two verify voltages, verify low VL and verify high VH. If a memory cell is detected in a program loop to have a threshold voltage Vt between verify low VL and verify high VH, then in the following VPGM pulse, a QPW bias voltage is applied to the bit line coupled with that memory cell. The QPW bias voltage increases the voltage in the channel containing the memory cell for which slow programming is desired, thereby reducing the voltage difference between the VPGM voltage and the channel and slowing the flow of electrons into the charge trapping material of the memory cell being programmed. Any memory cell with a threshold voltage VT that passes the verify high VH voltage is still locked down, or inhibited, for all remaining program loops by applying an inhibit voltage to the bit line coupled with the memory cell during the VPGM pulse and by skipping verify for that memory cell.

Typically, a programming operation continues uninterruptedly from beginning until end as illustrated in the example timeline depicted in FIG. **11** where "Prog" depicts a VPGM pulse and "V**1**" and "V**2**" depict verify pulses for any two data states and/or a verify low and a verify high for the same data state (additional verify pulses could also be conducted in each program loop). However, in some cases, a programming operation can be suspended so that the memory device can do another operation, e.g., read data in another memory block. The "delay time" (the time that the programming operation of the selected word line WLn is suspended) and the "suspend duration" (the duration of the suspension) can vary depending on the other operation that is performed. For example, in some cases, the suspend duration can be very short (e.g., less than 1 ms), very long (e.g., more than 10 s), or anywhere in between. A suspend-resume-go (SRG) technique is a mode where the memory device pauses the programming operation at the delay time for the suspend duration and resumes the programming operation once the suspend duration is completed. In most cases, the suspend duration is unknown at the delay time.

In some cases, during the suspend duration, the threshold voltages Vt of the memory cells in the selected word line can begin to fall due to short term data retention (STDR) effects. This phenomenon is sometimes known as "Vt drift" and can result in either over-programming or under-programming after the programming operation resumes depending on a number of factors including the delay time and the suspend duration. More specifically, in some cases, it has been found that over-programming occurs when the suspend duration is greater than fifty milliseconds (t_Suspend>50 ms), and under-programming occurs when the delay time is greater than seven hundred microseconds (t_Delay>700 μs) and the suspend duration is less than fifty milliseconds (t_Suspend<50 ms).

FIG. **12** illustrates the path the threshold voltage Vt of a single memory cell of a selected word line during such a programming operation that gets suspended and where over-programming occurs. In this figure, line **1200** illustrates the increase in the threshold voltage Vt of an example memory cell during a first VPGM pulse, which may or may not be a first VPGM pulse of the programming operation. Following the first VPGM pulse, the threshold voltage Vt remains below the verify low VL voltage, and thus, in the second VPGM pulse, the memory cell receives another VPGM pulse without the QPW bias to slow programming. Line **1202** illustrates the increase in the threshold voltage Vt during the second VPGM pulse. At this point in time (after the second VPGM pulse and before the third VPGM pulse), the programming operation is suspended. Vt drift causes the threshold voltage Vt of the memory cell to fall from between the verify low VL voltage and the verify high VH voltage to back below the verify low VL voltage, as illustrated with line **1204**. Programming then resumes when the suspension event is completed. As illustrated with line **1206**, in the third VPGM pulse, the threshold voltage Vt of this memory cell is increased to nearly the verify high VH voltage. Finally, in the fourth VPGM pulse, the memory cell is programmed at a slower rate due to the application of a QPW bias voltage to the bit line coupled to the memory cell and the threshold voltage Vt is increased (line **1208**) to its final threshold voltage Vt, which is higher than it would have been if not for the suspension of the programming operation. This may happen to other memory cells in the selected word line, and thus, the final threshold voltage distribution (line **1210**) for the target data state is wider (less desirable) because of the suspension than it would have been if there was no suspension (line **1214**). This results in a reduced Vt window and less data reliability.

FIG. **13** illustrates the path the threshold voltage Vt of a single memory cell of a selected word line during such a programming operation that gets suspended and where under-programming occurs. In this figure, line **1300** illustrates the increase in the threshold voltage Vt of an example memory cell during a first VPGM pulse, which may or may not be a first VPGM pulse of the programming operation. Following the first VPGM pulse, the threshold voltage Vt remains below the verify low VL voltage, and thus, in the second VPGM pulse, the memory cell receives another VPGM pulse without the QPW bias to slow programming. Line **1302** illustrates the increase in the threshold voltage Vt during the second VPGM pulse. At this point in time (after the second VPGM pulse and before the third VPGM pulse), the programming operation is suspended. In this case, the Vt drift is small and the threshold voltage Vt of the memory cell remains above the verify low VL voltage, as illustrated with line **1304**. Programming then resumes when the suspension event is completed. As illustrated with line **1306**, the third VPGM pulse is applied along with a QPW bias such that the final threshold voltage Vt of the memory cell is lower than it would have been if no suspension had occurred. This pulls down an upper tail of the final threshold voltage Vt distribution (line **1308**) for this data state as compared to if no suspension event had occurred (line **1310**).

FIG. **14** is a schematic view of an example memory cell during a suspension event. The memory cell includes a control gate **1400**, a blocking oxide layer **1402**, a charge trapping layer **1404**, and a tunnel oxide layer **1406**. As illustrated, at this time, electrons have amassed in the tunnel oxide layer **1406** at a very shallow location that is distant from the charge trapping layer **1404**. During a typical suspension event, some of those electrons at the shallow traps can leak out of the tunnel oxide layer **1406** and into the channel **1408**. This causes the STDR, or threshold voltage drive (Vt drift), which can lead to the above-discussed under-programming and over-programming issues once programming resumes. However, according to the embodiments of the subject disclosure, by maintaining a voltage at the control gate **1400** during the suspension event, these electrons are held in the memory cell and are restricted from leaking out into the channel **1408**.

The subject disclosure is related to a programming technique to reduce STDR and alleviate both under-programming and over-programming. According to these techniques, in the event of a suspension event, a charge is applied to the control gate **1400** of a selected word line WLn and is held at the control gate **1400** during the suspension event. By holding a charge at the control gate **1400** of the selected word line, the electrons in the tunnel oxide layer **1406** are substantially restricted or prevented from detrapping into the channel **1408**. Thus, Vt drift is minimized and, when the suspension event ends, programming can continue as if no suspension event had occurred. In other words, the next program loop can proceed the same as it would have if the programming operation had not been suspended at all.

FIG. **15** is a schematic view of an exemplary NAND string that includes a plurality of memory cells that are arranged in a plurality of word lines. Each of the word lines is in electrical communication with a respective word line driver (WL Driver), which is located in a CMOS layer that is attached with the array. A word line switch (WLSW) is electrically disposed between each WL Driver and the associated word line. The WLSW can be either turned on (made conductive to electrons) by applying a high voltage to its gate or turned off (made non-conductive to electrons) by removing the voltage from its gate. When the WLSW is turned on, the WL driver and the associated the word line are electrically connected with one such that a voltage change at the WL driver will correspondingly alter the voltage at the associated word line. When the WLSW is turned off, the word line is electrically isolated from its associated WL driver such that a change in voltage at the WL driver will not alter the voltage at the associated word line.

Turning now to FIG. **16**, an exemplary embodiment of the voltages applied to the WL Driver associated with a selected word line WLn, the WLSW associated with the selected word line WLn, and the selected word line WLn itself during a SRG operation according to an exemplary embodiment are depicted. As illustrated, at the time an instruction to suspend is received, the WLSW being "turned on" and the WL Driver and the selected word line WLn are at an elevated voltage, e.g., a programming voltage VPGM or a reference voltage VCG. At time t**1**, the WLSW is left on and the voltage of the WL Driver is ramped down from the elevated voltage to a reduced gate holding voltage PSRG. In an exemplary embodiment, the gate holding voltage PSRG is in the range of two to eight Volts (2-8 V). Because the WLSW is left on, ramping down the WL driver causes the selected word line WLn to also ramp down from the elevated voltage to the reduced voltage PSRG. Then, at time t**2**, the WLSW is turned off by cutting the voltage applied to its control gate, thereby electrically isolating the selected word line WLn from the WL Driver. At time t**3**, which can occur shortly after time t**2**, the WL Driver is ramped down from the gate holding voltage PSRG to approximately zero Volts (0 V). Because it is electrically isolated from the WL driver at time t**3**, the selected word line WLn remains at the reduced voltage PSRG throughout the suspension duration. By holding the selected word line WLn at the reduced voltage PSRG, detrappment of the electrons in the tunnel oxide layer is reduced and STDR is minimized.

Turning now to FIG. **17**, a flow chart **1700** is provided depicting the steps of performing a suspend-resume-go operation according to an exemplary embodiment of the present disclosure. These steps could be performed by the controller; a processor or processing device or any other circuitry, executing instructions stored in memory; and/or other circuitry described herein that is specifically configured/programmed to execute the following steps.

At step **1702**, a programming operation is occurring when a suspension event instruction is received. This could occur during, for example, a VPGM pulse or during a program-verify pulse. At step **1704**, the WL driver that is electrically connected to the selected word line WLn is ramped down from its previous voltage (e.g., an elevated voltage such as VPGM or VCG) to a reduced gate holding voltage PSRG. At step **1706**, the WLSW that is electrically connected between the WL Driver and the selected word line WLn is turned off to electrically isolate the selected word line WLn from the associated WL Driver. At step **1708**, which occurs after step **1706**, the WL Driver is ramped down to approximately zero Volts (0 V). Because the selected word line WLn is electrically isolated from the WL Driver at this time, the selected word line WLn remains at the gate holding voltage PSRG for the entire suspension event. Accordingly, the memory cells of the selected word line WLn experience very little STDR during the suspension event. When the suspension event is over, the programming operation resumes where it left off.

Various terms are used herein to refer to particular system components. Different companies may refer to a same or similar component by different names and this description does not intend to distinguish between components that differ in name but not in function. To the extent that various functional units described in the following disclosure are referred to as "modules," such a characterization is intended to not unduly restrict the range of potential implementation mechanisms. For example, a "module" could be implemented as a hardware circuit that includes customized very-large-scale integration (VLSI) circuits or gate arrays, or off-the-shelf semiconductors that include logic chips, transistors, or other discrete components. In a further example, a module may also be implemented in a programmable hardware device such as a field programmable gate array (FPGA), programmable array logic, a programmable logic device, or the like. Furthermore, a module may also, at least in part, be implemented by software executed by various types of processors. For example, a module may comprise a segment of executable code constituting one or more physical or logical blocks of computer instructions that translate into an object, process, or function. Also, it is not required that the executable portions of such a module be physically located together, but rather, may comprise disparate instructions that are stored in different locations and which, when executed together, comprise the identified module and achieve the stated purpose of that module. The executable code may comprise just a single instruction or a set of multiple instructions, as well as be distributed over different code segments, or among different programs, or across several memory devices, etc. In a software, or partial software, module implementation, the software portions may be stored on one or more computer-readable and/or executable storage media that include, but are not limited to, an electronic, magnetic, optical, electromagnetic, infrared, or semiconductor-based system, apparatus, or device, or any suitable combination thereof. In general, for purposes of the present disclosure, a computer-readable and/or executable storage medium may be comprised of any tangible and/or non-transitory medium that is capable of containing and/or storing a program for use by or in connection with an instruction execution system, apparatus, processor, or device.

Similarly, for the purposes of the present disclosure, the term "component" may be comprised of any tangible, physical, and non-transitory device. For example, a component may be in the form of a hardware logic circuit that is comprised of customized VLSI circuits, gate arrays, or other integrated circuits, or is comprised of off-the-shelf semiconductors that include logic chips, transistors, or other discrete components, or any other suitable mechanical and/or electronic devices. In addition, a component could also be implemented in programmable hardware devices such as field programmable gate arrays (FPGA), programmable array logic, programmable logic devices, etc. Furthermore, a component may be comprised of one or more silicon-based integrated circuit devices, such as chips, die, die planes, and packages, or other discrete electrical devices, in an electrical communication configuration with one or more other components via electrical conductors of, for example, a printed circuit board (PCB) or the like. Accordingly, a module, as defined above, may in certain embodiments, be embodied by or implemented as a component and, in some instances, the terms module and component may be used interchangeably.

Where the term "circuit" is used herein, it includes one or more electrical and/or electronic components that constitute one or more conductive pathways that allow for electrical current to flow. A circuit may be in the form of a closed-loop configuration or an open-loop configuration. In a closed-loop configuration, the circuit components may provide a return pathway for the electrical current. By contrast, in an open-looped configuration, the circuit components therein may still be regarded as forming a circuit despite not including a return pathway for the electrical current. For example, an integrated circuit is referred to as a circuit irrespective of whether the integrated circuit is coupled to ground (as a return pathway for the electrical current) or not. In certain exemplary embodiments, a circuit may comprise a set of integrated circuits, a sole integrated circuit, or a portion of an integrated circuit. For example, a circuit may include customized VLSI circuits, gate arrays, logic circuits, and/or other forms of integrated circuits, as well as may include off-the-shelf semiconductors such as logic chips, transistors, or other discrete devices. In a further example, a circuit may comprise one or more silicon-based integrated circuit devices, such as chips, die, die planes, and packages, or other discrete electrical devices, in an electrical communication configuration with one or more other components via electrical conductors of, for example, a printed circuit board (PCB). A circuit could also be implemented as a synthesized circuit with respect to a programmable hardware device such as a field programmable gate array (FPGA), programmable array logic, and/or programmable logic devices, etc. In other exemplary embodiments, a circuit may comprise a network of non-integrated electrical and/or electronic components (with or without integrated circuit devices). Accordingly, a module, as defined above, may in certain embodiments, be embodied by or implemented as a circuit.

It will be appreciated that example embodiments that are disclosed herein may be comprised of one or more microprocessors and particular stored computer program instructions that control the one or more microprocessors to implement, in conjunction with certain non-processor circuits and other elements, some, most, or all of the functions disclosed herein. Alternatively, some or all functions could be implemented by a state machine that has no stored program instructions, or in one or more application-specific integrated circuits (ASICs) or field-programmable gate arrays (FPGAs), in which each function or some combinations of certain of the functions are implemented as custom logic. A combination of these approaches may also be used. Further, references below to a "controller" shall be defined as comprising individual circuit components, an application-specific integrated circuit (ASIC), a microcontroller with controlling software, a digital signal processor (DSP), a field programmable gate array (FPGA), and/or a processor with controlling software, or combinations thereof.

Additionally, the terms "couple," "coupled," or "couples," where may be used herein, are intended to mean either a direct or an indirect connection. Thus, if a first device couples, or is coupled to, a second device, that connection may be by way of a direct connection or through an indirect connection via other devices (or components) and connections.

Regarding, the use herein of terms such as "an embodiment," "one embodiment," an "exemplary embodiment," a "particular embodiment," or other similar terminology, these terms are intended to indicate that a specific feature, structure, function, operation, or characteristic described in connection with the embodiment is found in at least one embodiment of the present disclosure. Therefore, the appearances of phrases such as "in one embodiment," "in an embodiment," "in an exemplary embodiment," etc., may, but do not necessarily, all refer to the same embodiment, but rather, mean "one or more but not all embodiments" unless expressly specified otherwise. Further, the terms "comprising," "having," "including," and variations thereof, are used in an open-ended manner and, therefore, should be interpreted to mean "including, but not limited to . . . " unless expressly specified otherwise. Also, an element that is preceded by "comprises . . . a" does not, without more constraints, preclude the existence of additional identical elements in the subject process, method, system, article, or apparatus that includes the element.

The terms "a," "an," and "the" also refer to "one or more" unless expressly specified otherwise. By way of example, "a processor" programmed to perform various functions refers to one processor programmed to perform each and every function or more than one processor collectively programmed to perform each of the various functions. In addition, the phrase "at least one of A and B" as may be used herein and/or in the following claims, whereby A and B are variables indicating a particular object or attribute, indicates a choice of A or B, or both A and B, similar to the phrase "and/or." Where more than two variables are present in such a phrase, this phrase is hereby defined as including only one of the variables, any one of the variables, any combination (or sub-combination) of any of the variables, and all of the variables.

Further, where used herein, the term "about" or "approximately" applies to all numeric values, whether or not explicitly indicated. These terms generally refer to a range of numeric values that one of skill in the art would consider equivalent to the recited values (e.g., having the same function or result). In certain instances, these terms may include numeric values that are rounded to the nearest significant figure.

In addition, any enumerated listing of items that is set forth herein does not imply that any or all of the items listed are mutually exclusive and/or mutually inclusive of one another, unless expressly specified otherwise. Further, the term "set," as used herein, shall be interpreted to mean "one or more," and in the case of "sets," shall be interpreted to mean multiples of (or a plurality of) "one or more," "ones or more," and/or "ones or mores" according to set theory, unless expressly specified otherwise.

The foregoing detailed description has been presented for purposes of illustration and description. It is not intended to be exhaustive or be limited to the precise form disclosed. Many modifications and variations are possible in light of the above description. The described embodiments were chosen to best explain the principles of the technology and its practical application to thereby enable others skilled in the art to best utilize the technology in various embodiments and with various modifications as are suited to the particular use contemplated. The scope of the technology is defined by the claims appended hereto.

What is claimed is:

1. A method of performing a programming operation in a memory device, comprising the steps of:

preparing a memory block that includes an array of memory cells that are arranged in a plurality of word lines, the word lines being in electrical communication with respective word line drivers and word line switches;

in a program loop of a programming operation, applying an elevated voltage to a selected word line of the plurality of word lines;

receiving a command to suspend the programming operation; and

with the word line switch associated with the selected word line turned on, ramping the selected word line from the elevated voltage to a reduced gate holding voltage and then turning off the word line switch associated with the selected word line to electrically isolate the selected word line from the associated word line driver so that the selected word line remains at the gate holding voltage until the programming operation resumes.

2. The method as set forth in claim **1**, wherein the elevated voltage is a programming voltage.

3. The method as set forth in claim **1**, wherein the elevated voltage is a reference voltage that is associated with one of a plurality of programmed data states that the memory cells of the selected word line are being programmed to.

4. The method as set forth in claim **1**, wherein the gate holding voltage is in the range of two to eight Volts.

5. The method as set forth in claim **1**, wherein after the step of turning the word line switch associated with the selected word line off, the method further includes the step of ramping down the word line driver associated with the selected word line from the gate holding voltage to approximately zero Volts while the selected word line remains at the gate holding voltage.

6. The method as set forth in claim **1**, wherein during the programming operation, the memory cells of the selected word line are being programmed to at least three bits of data per memory cell.

7. The method as set forth in claim **1**, wherein the plurality of word lines further includes a plurality of unselected word lines, and following the step of receiving the instruction to suspend the programming operation, the method further includes the step of ramping the word line drivers that are associated with the unselected word lines directly to approximately zero Volts.

8. A memory device, comprising:

a memory block that includes an array of memory cells that are arranged in a plurality of word lines, the word lines being in electrical communication with respective word line drivers and word line switches;

control circuitry that is configured to program the memory cells of a selected word line of the plurality of word lines in a programming operation, during the programming operation, the control circuitry being configured to;

apply an elevated voltage to the selected word line;

receive a command to suspend the programming operation; and

with the word line switch associated with the selected word line turned on, ramp the selected word line from the elevated voltage to a reduced gate holding voltage and then turn the word line switch associated with the selected word line off to electrically isolate the selected word line from the associated word line driver so that the selected word line remains at the gate holding voltage until the programming operation resumes.

9. The memory device as set forth in claim **8**, wherein the elevated voltage is a programming voltage.

10. The memory device as set forth in claim **8**, wherein the elevated voltage is a reference voltage that is associated with one of a plurality of programmed data states that the memory cells of the selected word line are being programmed to.

11. The memory device as set forth in claim **8**, wherein the gate holding voltage is in the range of two to eight Volts.

12. The memory device as set forth in claim **8**, wherein after turning off the word line switch associated with the selected word line, the method further includes the step of ramping down the word line driver associated with the selected word line from the gate holding voltage to approximately zero Volts while the selected word line remains at the gate holding voltage.

13. The memory device as set forth in claim **8**, wherein during the programming operation, the control circuitry is programming the memory cells of the selected word line to at least three bits of data per memory cell.

14. The memory device as set forth in claim **8**, wherein the plurality of word lines further includes a plurality of unselected word lines, and following the control circuitry receiving the instruction to suspend the programming operation, the control circuitry ramps the word line drivers that are associated with the unselected word lines directly to approximately zero Volts.

15. An apparatus, comprising:

a memory block that includes an array of memory cells that are arranged in a plurality of word lines, the word lines being in electrical communication with respective word line drivers and word line switches;

a programming means for programming the memory cells of a selected word line of the plurality of word lines in a programming operation, during the programming operation, the programming means being configured to;

apply an elevated voltage to the selected word line;

receive a command to suspend the programming operation; and

with the word line switch associated with the selected word line turned on, ramp the selected word line from the elevated voltage to a reduced gate holding voltage and then turn the word line switch associated with the selected word line off to electrically isolate the selected word line from the associated word line driver so that the selected word line remains at the gate holding voltage until the programming operation resumes.

16. The apparatus as set forth in claim **15**, wherein the elevated voltage is a programming voltage.

17. The apparatus as set forth in claim **15**, wherein the elevated voltage is a reference voltage that is associated with one of a plurality of programmed data states that the memory cells of the selected word line are being programmed to.

18. The apparatus as set forth in claim **15**, wherein the gate holding voltage is in the range of two to eight Volts.

19. The apparatus as set forth in claim **15**, wherein after turning off the word line switch associated with the selected word line, the programming means ramps down the word line driver associated with the selected word line from the gate holding voltage to approximately zero Volts while the selected word line remains at the gate holding voltage.

20. The apparatus as set forth in claim **15**, wherein during the programming operation, the programming means is programming the memory cells of the selected word line to at least three bits of data per memory cell.

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